

Form 63D

(December 2006)

ACT Assessment®

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In response to your recent request for test information release materials, this booklet contains the test questions and conversion tables used in determining your ACT scores. Enclosed with this booklet is a report listing your answers to the ACT Assessment tests and the answer key.

If you wish to order a photocopy of your answer document—including, if you took the Writing Test, a copy of your written essay—please use the order form on the inside back cover of this booklet.

We hope that you will find this information helpful.

1

ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose "NO CHANGE." In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Honeybee Love

Over the years, I have gained a reputation¹

that among family and friends as a honeybee enthusiast.²

My fascination with these little creatures³ puts me into some interesting situations now and then. Just last month, for instance, I received a call from a neighbor her plea⁴ for my help resulted in a happy ending for everyone involved.

Clara, my neighbor, explained her problem.

While she had been away for the summer, bees had invaded her house. She was pretty sure they had found their way in through a tiny hole near the chimney.

1. Which of the following alternatives to the underlined portion would NOT be acceptable?
 - A. developed
 - B. earned
 - C. promoted
 - D. acquired
2. F. NO CHANGE
 G. which is
 H. that is
 J. OMIT the underlined portion.
3. A. NO CHANGE
 B. fascination with these little creatures
 C. fascination, with these little creatures
 D. fascination, with these little creatures,
4. F. NO CHANGE
 G. whose
 H. a
 J. expressing a

1 1 1 1 1 1 1 1 1 1

In the hollow space between the outside and inside walls,
the bees had established a hive, much to Clara's alarm. 5

She could now hear, the thrum and whir of the colony

sitting in her living room. Outside, she could see the
bees coming and going from her house, which they
obviously treated as theirs, too.

Clara asked me to remove the bees. I explained that
I was knowledgeable about bees and enchanted by their
behavior, but I had no beekeeping expertise. However,
I assured her, I would find someone who did have some.
In the meantime, I told her, I would be happy to come
by and size up to her situation.

When the time came, Clara wanted to know if the

bees in her house were dangerous, I assured her that
they were docile honeybees. Hoping to cheer her up,
I pointed out that if my estimates were correct the hive
might produce as much as two hundred pounds of honey
a year for her. She still wanted the bees banished. I could
have them and their honey too as far as she was concerned.

5. The writer is considering deleting the preceding sentence from the essay. Should the sentence be kept or deleted?

- A. Kept, because it is relevant to the description of how the bees became a problem.
- B. Kept, because it establishes where the narrator reached in to retrieve the bees.
- C. Deleted, because the information it contains is already provided in the preceding sentence.
- D. Deleted, because the story is more effective if Clara's alarm is not revealed until later.

6. F. NO CHANGE

- G. hear the thrum, and whir
- H. hear the thrum and whir,
- J. hear the thrum and whir

7. A. NO CHANGE

- B. as she sat
- C. seated
- D. being seated

8. F. NO CHANGE

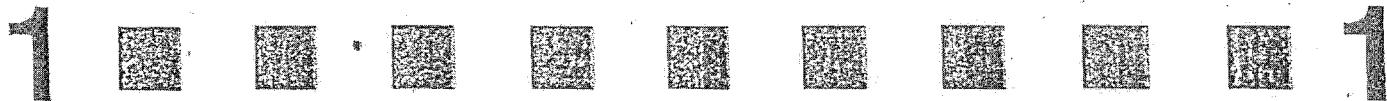
- G. up
- H. the extent of
- J. OMIT the underlined portion.

9. Given that all the choices are true, which one most strongly reinforces Clara's attitude toward the bees as it has been conveyed up to this point in the essay?

- A. NO CHANGE
- B. As soon as I arrived,
- C. Having given the matter some thought,
- D. During my visit, I learned that

10. F. NO CHANGE

- G. dangerous, my comments were intended to assure
- H. dangerous, I did my best to assure
- J. dangerous. I assured



So, instead, I made the leap from bee-lover to beekeeper.

On a website devoted to bees, I found the name of a

beekeeper in our region. ¹¹ She was eager to help out even though she lived an hour away. One Saturday

afternoon, ¹³ her and I transferred the bees—and their honey—to a wooden hive. With her help, I am learning to tend it. Clara is relieved, I'm happy, the bees are settled, and my family and friends are looking forward to the honey. ¹⁴

11. A. NO CHANGE
B. (Do NOT begin new paragraph) So, after it was all over,
C. (Begin new paragraph) So
D. (Begin new paragraph) For instance,
12. At this point, the writer is considering adding the following true statement:
I have found very useful information on the Internet regarding how bees communicate with each other.
Should the writer make this addition here?
F. Yes, because it helps support the idea that the narrator is fascinated with bees.
G. Yes, because it provides a strong contrast between the narrator's interests and Clara's.
H. No, because information regarding bee communication is also available in magazines and textbooks and on videos.
J. No, because it is not relevant to the narrative at this point in the essay.
13. A. NO CHANGE
B. she and I
C. me and her
D. she and me
14. The writer is considering concluding the essay with the following statement:
Once again, necessity is the mother of invention.
Should the writer end the essay with this statement?
F. Yes, because it sums up the events of the essay in a memorable way.
G. Yes, because Clara's request for help adds to the central suspense of the essay.
H. No, because the preceding sentence contains the same meaning expressed in different words.
J. No, because it does not have a meaningful connection to the topic of this essay.

Question 15 asks about the preceding passage as a whole.

15. Suppose that one of the writer's goals had been to indicate how two people viewed the same situation differently. Would this essay fulfill that goal?
A. Yes, because the essay reveals that the narrator and Clara have contrasting reactions to the bees' occupying Clara's house.
B. Yes, because the essay reveals that Clara thinks the bees can't be removed without someone being hurt, but the narrator believes they can.
C. No, because the essay indicates that the narrator understands why Clara is afraid even though the narrator does not share her fear.
D. No, because the essay establishes that neither Clara nor the narrator has as much experience with bees as the beekeeper does.

PASSAGE II

A Photographer Discovered

In the early twentieth century, as cameras became easier to use and more affordable, amateur photographers began capturing scenes and subjects that had previously been neglected. One of these ¹⁶ photographers, Earl McWilliams, was active in Lincoln, Nebraska, between 1910 and 1920. Recently, about 300 of his negatives were discovered. Art critics and historians consider them significant for several reasons.

First, since they came from the past, ¹⁷

McWilliams's photographs in a small town provide ¹⁸ a rare and valuable record of the daily life of African

Americans in the early 1900s. Although ¹⁹ photographs of African American life in major cities like Chicago and New York are not uncommon, few records of life in towns like Lincoln have survived.

McWilliams's photographs also demonstrate a fine sense of composition—an ability to arrange the subject of the photograph within the surrounding objects. In one portrait, a thoughtful-looking woman stands on a large porch. The porch's ²⁰ railings and columns frame her on one side, and the house does so on the other. These framing lines give the photograph a sense of balance.

16. F. NO CHANGE
G. have previously been
H. are previously being
J. are previously

17. A. NO CHANGE
B. being that they have preserved the past,
C. because they are from a long time ago,
D. OMIT the underlined portion.

18. The best placement for the underlined portion would be:
F. where it is now.
G. after the word *provide*.
H. after the word *record*.
J. after the word *Americans*.

19. Which of the following alternatives to the underlined portion would be LEAST acceptable?
A. Whether
B. Whereas
C. Though
D. While

20. F. NO CHANGE
G. porches'
H. porches
J. porch,

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Furthermore, the uncluttered background directs
the viewer's eye to the woman herself as she
looks off into the distance as if one is waiting
21

for someone. 22

It is suggested that McWilliams's photographs were
23
also gifted with the ability to make people feel comfortable
in front of the camera. One picture shows a little
boy standing on a platform. On either side
of him is a dog that the boy holds the dogs'
24
collars in his hands. He smiles at the camera,
and one has to wonder what made him happy.
25
One art critic commented that it appears as if

boy and dogs alike found it completely natural
26
to stop and pose for McWilliams.

Little is known of Earl McWilliams
accept that he worked as an elevator operator at
27

a department store and later as a photofinisher.
28

21. A. NO CHANGE
B. they were waiting
C. she waited
D. waiting
22. At this point, the writer is thinking of adding the following sentence:
Of course, we will never know who that someone is or when he or she will appear.
Should the writer make this addition here?
F. Yes, because it leads the reader to identify with the subject of the photograph.
G. Yes, because it emphasizes the point about how old McWilliams's photographs are.
H. No, because it strays too far from the paragraph's focus on McWilliams's skills as a photographer.
J. No, because it encourages the reader to imagine the thoughts of the woman in the photograph.
23. Which choice provides the most concise and stylistically effective wording here?
A. NO CHANGE
B. There is the suggestion in McWilliams's photographs that he was
C. It is suggested by McWilliams's photographs that he was
D. McWilliams's photographs suggest that he was
24. F. NO CHANGE
G. dog that the boy is holding
H. dog; the boy holds
J. dog the boy holds
25. Given that all the choices are true, which one would best serve the purpose of further describing this photograph?
A. NO CHANGE
B. which produced plate glass negatives.
C. and the dogs seem to be wagging their tails.
D. a cumbersome and inexpensive box camera.
26. F. NO CHANGE
G. to be their usual and customary practice
H. to be their accustomed activity
J. totally free and easy
27. A. NO CHANGE
B. except that
C. except for
D. accepting
28. F. NO CHANGE
G. working as a photofinisher.
H. as a photofinishing job.
J. doing photofinishing.

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Fortunately, he spends his spare time among Lincoln's

29

residents, and he may soon gain national acclaim.

30

29. A. NO CHANGE

B. is spending

C. spent

D. has spent

30. Given that all the choices are true, which one would best conclude this essay by summarizing one of its main points?

F. NO CHANGE

G. capturing moments on film that remain vivid today.

H. carrying around his cumbersome camera equipment.

J. when he wasn't working at his regular job.

PASSAGE III

A Stormy Ride

[1] When I moved from New York City to rural

Nebraska, I became friends with a rancher, named Amil.

31

www.crackab.com

[2] For his livelihood, he raised cattle for a living, but he

32

enjoyed most the herd of tall, magnificence horses he kept.

33

[3] I had no experience with horses when I had always

34

been too scared to ride them. [4] I was reluctant but finally

agreed to give it a try. [5] One day, though, Amil asked

35

me if I would like to sit in the saddle for a while. 36

Amil brought Stormy and Morgan, two enormous mares, out of the stable. When all was ready, he told me to get on them. I held the reins, put my foot in the stirrup, and hoisted myself up into the saddle. I sat there, awkwardly holding the saddle horn as if it were a hand brake.

31. A. NO CHANGE

B. rancher; named

C. rancher named

D. rancher named,

32. F. NO CHANGE

G. He had started the raising of

H. His livelihood was raising

J. He raised

33. A. NO CHANGE

B. tall; magnificent

C. magnificently, tall

D. tall magnificence

34. F. NO CHANGE

G. horses, which

H. horses;

J. horses, however,

35. A. NO CHANGE

B. day though:

C. day, though

D. day, though:

36. For the sake of the logic and coherence of this paragraph, Sentence 4 should be placed:

F. where it is now.

G. before Sentence 1.

H. after Sentence 2.

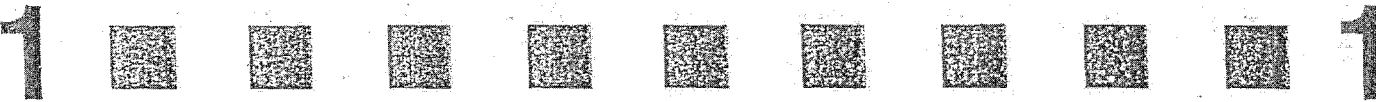
J. after Sentence 5.

37. A. NO CHANGE

B. her.

C. Stormy.

D. the horse.



I had followed Amil's initial instructions only to realize he had not told me what to do next. I didn't even know how to get Stormy to start moving.

Amil had a good-natured smile on his face, I think, he could tell how nervous I was. Sitting at the controls of a jumbo jet, I might just as well have been flying. I watched

him gently nudge his heels into Morgan's side, which she quickly responded and trotted down the path.

Stormy must have sensed that I was a novice. Suddenly, on her own, she started to walk toward a field of flowing grass. At first, I was

alarmed like a distress signal flashing in my mind, but she was moving slowly, and I realized that I would of been able to hang on. Then, after about a minute, she stopped walking, bent her head, and calmly started eating.

"Well," Amil said as he rode by, "you've just led Stormy to her favorite patch of grass. And I thought you said you'd never ridden before."

So there I sat, atop a beautiful, massive horse, and watched as she ate her dinner. I wasn't an expert rider, but at least I was an easy one to get along with. Although I later learned how to really ride a horse, I'll never forget my first time, when Stormy took me for a short, gentle ride.

- 38. F. NO CHANGE
G. Amils' initial instructions,
H. Amils initial instructions
J. Amils' initial instructions
- 39. A. NO CHANGE
B. face I think,
C. face. I think
D. face, however, I think
- 40. F. NO CHANGE
G. I might just as well have been sitting at the controls of a jumbo jet.
H. Amil could tell I was nervous as if I was flying, sitting at the controls of a jumbo jet.
J. I might, sitting at the controls of a jumbo jet, have been just as nervous as if I was flying.
- 41. A. NO CHANGE
B. side, which it
C. side. She
D. side, where she
- 42. Given that all of the choices are true, which one most effectively introduces the action in this paragraph while suggesting the narrator's inability to control Stormy?
F. NO CHANGE
G. Amil really knew how to ride!
H. Though very tall, Stormy was a gentle creature.
J. Amil had bought Stormy from another rancher some years ago.
- 43. A. NO CHANGE
B. alarmed with fear and distress,
C. alarmed by the immediate awareness of danger,
D. alarmed,
- 44. F. NO CHANGE
G. would be
H. will be
J. should of been
- 45. A. NO CHANGE
B. a rider who rode with great skill or expertise,
C. a master of the art of horseback riding,
D. an expert,

PASSAGE IV

Marine Entomology—The Study of Ocean Insects

Scientific curiosity is often motivated by observable phenomena. Sometimes though, as with the research of entomologist Dr. Lanna Cheng, that curiosity is fueled by what seem to be absent. Her interest in a curious paradox in nature has stimulated the growth of a new branch of oceanography.

Early in her career, Dr. Cheng became intrigued by the fact that because insects are the earth's most abundant and diverse organisms, they are virtually absent from and nonexistent in the world's largest habitat,

the ocean. To try to solve this riddle, Dr. Cheng began studying the genus *Halobates*, the only known open-ocean

insects. She became the first scientist to capture

these insects alive and bringing them back to the laboratory for study.

46. F. NO CHANGE
G. seems
H. seemed
J. have seemed

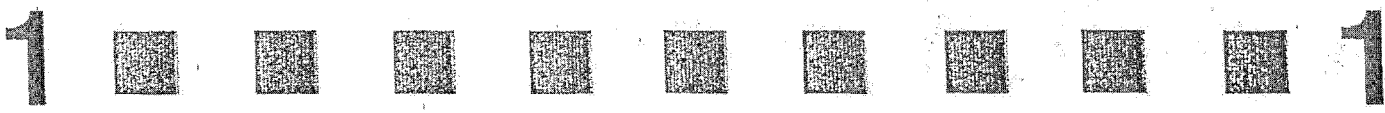
47. A. NO CHANGE
B. though
C. since
D. OMIT the underlined portion.

48. F. NO CHANGE
G. absent from
H. not only absent from but also nonexistent in
J. absent and not present in

49. Given that all the choices are true, which one provides the most effective transition from the first part of this paragraph to this sentence?
A. NO CHANGE
B. Using her maiden name, not her married name,
C. As a result of a scientific report,
D. During a walk on the beach,

50. If the writer were to delete the phrase "the only known open-ocean insects" from the preceding sentence (placing a period after the word *Halobates*), the paragraph would primarily lose:
F. information that helps suggest why insects are so abundant and diverse.
G. a description that explains why these insects are being studied.
H. a reference that acknowledges the many unknown insects.
J. an unnecessary detail, because this information is provided earlier in the essay.

51. A. NO CHANGE
B. will bring
C. bring
D. is bringing



Through her research on *Halobates* and other insects, Dr. Cheng has considered—and then dismissed—three hypotheses concerning the scarcity of marine insects. The first offering the⁵² oceans high, salt content as an explanation. But Dr. Cheng has found species of flies that can tolerate more than three times the salt content of the sea. The second hypothesis states that the ocean is too deep and underwater pressure too great for insect larvae to complete it's life cycle. Research has found, however, that some fly larvae can survive at greater depths than those of the deepest-diving marine mammals. A third hypothesis asserts that ocean water contains too little oxygen for⁵³ insects. Similarly, some fly larvae can survive months without oxygen, while other insect species live in very polluted waters with similarly low oxygen levels.

Dr. Cheng believes that a fourth hypothesis is the likely explanation. She suspects insects are less successful at sea because they so readily adapted to land. While the insects were colonizing land, the crustaceans, their main marine competitors, were effectively adapting to the saltwater environment. As partial evidence, Dr. Cheng cites the fact that *Halobates* spend their lives walking on top of the⁵⁴ water; thus eliminating the need to compete with the crustaceans below.⁵⁵

52. F. NO CHANGE
G. has been offered
H. offers
J. will offer

53. A. NO CHANGE
B. oceans high
C. ocean's high
D. ocean's high,

54. F. NO CHANGE
G. there
H. they're
J. their

55. A. NO CHANGE
B. too few
C. lesser
D. fewer

56. F. NO CHANGE
G. Therefore,
H. Yet,
J. So

57. A. NO CHANGE
B. in all probability the likely
C. likely the most probable
D. very probably the likely

58. Which of the following alternatives to the underlined portion would NOT be acceptable?
F. upon
G. top of
H. atop
J. on

59. A. NO CHANGE
B. water and thus
C. water, thus
D. water. Thus

Question 60 asks about the preceding passage as a whole.

60. Suppose the writer's goal had been to write a brief essay about how a scientist's curiosity can lead to research studies. Would this essay successfully fulfill that goal?
- F. Yes, because it shows how Dr. Cheng based her research on a question she had about marine insects.
 - G. Yes, because it describes how Dr. Cheng was able to develop and ask various hypothetical questions.
 - H. No, because it offers only partial evidence to support Dr. Cheng's hypotheses.
 - J. No, because it focuses on the species *Halobates* rather than on Dr. Cheng.

PASSAGE V

A Case for Noise Pollution Controls

[1] Talk to friends and neighbors, and you'll have a hard time finding anyone who claims to be in favor of air and water pollution.

[2] Our environment is a common resource, something we all enjoy during exciting weekends in the park.

61

[3] We don't tolerate those who exploit or abuse,
the public lands we use for recreation, the water
we drink, or the air we breathe. [4] Yet many
people, who feel strongly, about pollution of these
resources are strangely silent about protecting

63

61. Which choice would most effectively conclude this sentence so that it emphasizes citizens' involvement in preserving the environment?
- A. NO CHANGE
 - B. we all have a right to use and a responsibility to protect.
 - C. that reminds us of our relationship to the land and its contribution to our peace and well-being.
 - D. we all reflect upon and appreciate for the many benefits it provides.
62. F. NO CHANGE
G. who exploit, or abuse
H. who exploit or abuse
J. who, exploit or abuse,
63. A. NO CHANGE
B. people who feel strongly about pollution,
C. people, who feel strongly about pollution,
D. people who feel strongly about pollution

our environment from noise. ⁶⁴

Advocates supporting noise pollution controls are ⁶⁵ concerned about the noise from construction equipment and airplanes. They are also targeting the noise of snowmobiles and all-terrain vehicles, of leaf blowers ⁶⁶ and loud car sound systems. Some of the effects of excessive noise are already known. The most obvious one is premature hearing loss. Studies indicate a fairer ⁶⁷ clear

link between noise and stress, which ⁶⁸ experts believe there's a connection between noise and violence.

Therefore, ⁶⁹ there's evidence that noise affects not only humans but animals as well. For these reasons, there's increasing agreement with the view that ⁷⁰ "good neighbors keep their noise to themselves."

Of course, many people had insisted ⁷¹ that they have a right to make noise on their own property.

64. At this point, the writer is considering adding the following sentence:

These people have yet to understand the scope and effects of noise pollution.

Would this be a relevant addition to make here?

- F. Yes, because it completes the idea expressed in Sentence 4.
- G. Yes, because it identifies those who exploit and abuse the land referred to in Sentence 3.
- H. No, because it contradicts the idea expressed in Sentence 1.
- J. No, because it provides a digression that leads the essay away from its primary focus.

65. A. NO CHANGE
B. Advocates who are in favor of
C. Advocates of
D. People who advocate and support

66. F. NO CHANGE
G. vehicles;
H. vehicles and,
J. vehicles of

67. A. NO CHANGE
B. fair
C. more fairly
D. fairly

68. F. NO CHANGE
G. stress that
H. stress, and
J. stress because

69. A. NO CHANGE
B. Namely,
C. On the other hand,
D. In addition,

70. F. NO CHANGE
G. agreement increasing with the view
H. agreement with the view increasing
J. agreement with the increasing view

71. A. NO CHANGE
B. insisted
C. were insisting
D. will insist

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But do they? ⁷² Do people have a right to dump dangerous chemicals on their land if these chemicals escape into the air or end up in our rivers and reservoirs?

The laws say no. Many municipalities have ⁷³ begun to adopt ordinances that limit levels and kinds of noise.

These noise laws specify times when lawnmowers cannot be used, and they restrict when airplanes can take off and land. ⁷⁴ They crack down on loud car stereos,

just as they ban motorized watercraft. ⁷⁵ These steps suggest that even if the debate about noise controls may be loud, we may eventually end up with a quieter environment.

72. The writer is considering deleting the preceding sentence from this paragraph. Should the sentence be kept or deleted?

- F. Kept, because it provides a transition from the opening sentence to the statements at the end of this paragraph.
- G. Kept, because it is the first in a series of questions that explain the meaning of noise pollution.
- H. Deleted, because the question it poses is never answered in the paragraph.
- J. Deleted, because the question is restated in the next sentence and therefore is redundant.

73. A. NO CHANGE
B. began
C. beginning
D. begin

74. At this point, the writer is considering adding the following sentence:

Today airports can hardly control the number of airline arrivals and departures, and this air traffic congestion creates all sorts of other problems.

Should the writer add this sentence here?

- F. Yes, because it adds important details that suggest noise is not the only concern with major airports.
- G. Yes, because it provides additional information suggesting the extent to which noise laws have been enforced.
- H. No, because it does not mention the other problems that are occurring at major airports.
- J. No, because it provides a digression that leads the paragraph away from its primary focus.

75. The writer is considering adding the following phrase to the end of the preceding sentence:

from many lakes and rivers.

Should the writer add this phrase here?

- A. Yes, because it adds a specific detail clarifying how motorized watercraft have been restricted.
- B. Yes, because it supports the idea of public lands being used for recreation, mentioned in the first paragraph.
- C. No, because it does not specify from which lakes and rivers the watercraft have been banned.
- D. No, because it is redundant; the essay already implies that lakes and rivers would be the only waterways affected by loud noise.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

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(December 2006)

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MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. The Crossroads High School basketball team sponsored a student/faculty basketball game. Tickets sold for \$5.00 each. Each person with a ticket was entitled to admission to the game, 1 pretzel, and 1 can of soda. Each additional pretzel cost \$1.00 and each additional can of soda cost \$1.50. The basketball team sold 120 tickets. At the game, 36 additional pretzels were sold. The team collected a total of \$702.00. How many additional cans of soda were sold?

- A. 44
- B. 48
- C. 68
- D. 92
- E. 364

2. In football, total offense consists of rushing offense and passing offense. The table below shows rushing offense and total offense for a college football program during a 3-year period.

Rushing offense			
Year	Carries	Total yards	Yards per game
1997	397	1,028	93.5
1998	394	1,417	128.8
1999	378	1,920	213.3
Total offense			
Year	Plays	Total yards	Yards per game
1997	783	3,579	325.4
1998	759	3,692	335.6
1999	604	3,452	383.6

In 1997, what was the number of yards of total offense that was passing offense?

- F. 1,532
- G. 2,162
- H. 2,275
- J. 2,551
- K. 2,664

DO YOUR FIGURING HERE.



3. A 7-day week is how many times as long as 60 minutes?

A. 24
B. 31
C. 52
D. 84
E. 168

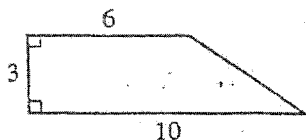
DO YOUR FIGURING HERE.

4. Veata earns f dollars per hour for h hours and then earns g dollars per hour for $3h$ hours. In terms of f , g , and h , these earnings totaled how many dollars?

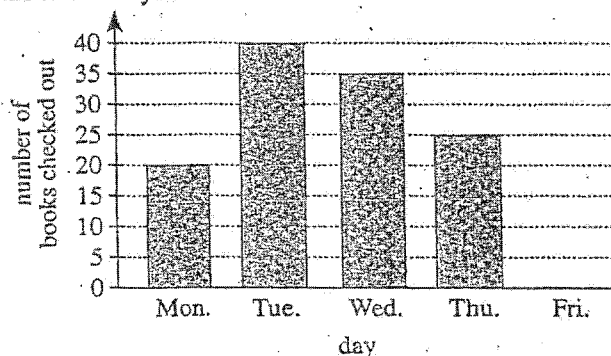
F. $3fgh$
G. $3fgh^2$
H. $fh + 3g$
J. $fh + 3gh$
K. $3fh + 3gh$

5. A trapezoid is shown below with the given dimensions in meters. What is the area, in square meters, of the trapezoid?

A. 9
B. 15
C. 18
D. 24
E. 48



6. The graph below shows the number of books checked out of the school library on the first 4 days of a 5-day week. How many books need to be checked out on Friday for the 5-day week's mean to equal the mean of the first 4 days?



F. 0
G. 24
H. 30
J. 40
K. 120

7. A ball is kicked upward. It is h feet above the ground t seconds after it is kicked. The relationship between h and t is given by the equation $h = 50t - 16t^2$. How many feet above the ground is the ball 2 seconds after it is kicked?

A. 34
B. 36
C. 50
D. 68
E. 84

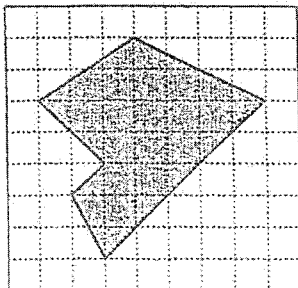


8. The high and low temperatures for a 5-day period are listed below. Which day had the greatest range of temperatures?

DO YOUR FIGURING HERE.

Day	High	Low
Monday	30°F	12°F
Tuesday	32°F	14°F
Wednesday	29°F	10°F
Thursday	25°F	8°F
Friday	28°F	16°F

- F. Monday
G. Tuesday
H. Wednesday
J. Thursday
K. Friday
9. Siblings Peter, Paul, and Mary earned a total of \$200 shoveling snow. If Peter earned 37% of the total and Paul earned \$16, what fraction of the \$200 did Mary earn?
- A. $\frac{1}{3}$
B. $\frac{1}{2}$
C. $\frac{11}{20}$
D. $\frac{131}{200}$
E. $\frac{147}{200}$
10. For $f(x) = 2x + 10$ and $g(x) = 3x^2$, what is the value of $f(g(-3))$?
- F. 31
G. 48
H. 64
J. 108
K. 144
11. Each small square on the grid shown below has a side length of 1 cm. Each vertex of the shaded region lies on a vertex of a small square. What is the area, in square centimeters, of the shaded region?



- A. 17
B. 23
C. 28
D. 29
E. 39



12. A roof rises 4 inches for each 12 inches of horizontal run. This roof rises $30\frac{1}{2}$ inches in how many inches of horizontal run?

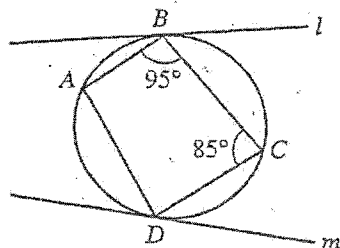
F. $10\frac{1}{6}$
 G. $22\frac{1}{2}$
 H. $38\frac{1}{2}$
 J. $91\frac{1}{2}$
 K. 122

DO YOUR FIGURING HERE.

13. When $a = 4$ and $b = -2$, $(2a - b) + b^a = ?$

A. -10
 B. -6
 C. 2
 D. 22
 E. 26

14. In the figure below, lines l and m are tangent to the circle at points B and D , respectively. Points A and C are on the circle. The measure of $\angle ABC$ is 95° and the measure of $\angle BCD$ is 85° .



The lines in which of the following pairs of lines are necessarily parallel?

- I. l and m
 II. $\overleftrightarrow{AB}$ and $\overleftrightarrow{DC}$
 III. $\overleftrightarrow{AD}$ and $\overleftrightarrow{BC}$

F. I only
 G. II only
 H. III only
 J. I and II only
 K. I, II, and III

15. Each day, a grocery store sells 112 gallons of milk for \$2.10 per gallon. The owner of the grocery store has determined that for each \$0.03 increase in the price per gallon of milk, the number of gallons sold will decrease by 2. If x represents the number of \$0.03 increases in the price of a gallon of milk, which of the following expressions represents the daily amount of money, in dollars, the grocery store will collect from the sale of milk?

A. $(2.10 - 0.03x)(112 - 2x)$
 B. $(2.10 + 0.03x)(112 - 2x)$
 C. $(2.10 + 0.03x)(112 - 0.02x)$
 D. $(2.10 - 3x)(112 - 2x)$
 E. $(2.10 + 3x)(112 - 2x)$

2



2

16. The expression $x^2 + 5x - 6$ is equivalent to:

DO YOUR FIGURING HERE.

- F. $(x - 6)(x + 1)$
- G. $(x - 6)(x + 5)$
- H. $(x - 3)(x - 2)$
- J. $(x - 3)(x + 2)$
- K. $(x - 1)(x + 6)$

17. When $a = 2$ and $b = 3$, the expression

$$\frac{ab}{25} + \frac{11}{10(a+b)} + \frac{1}{a+b} = ?$$

- A. $\frac{9}{40}$
- B. $\frac{11}{20}$
- C. $\frac{31}{50}$
- D. $\frac{47}{75}$
- E. $\frac{33}{50}$

www.crackab.com

18. The hours and minutes on a 12-hour digital clock are represented by 3 or 4 digits. Which of the following is the *largest* product that can be obtained by multiplying the digits in one of these representations?

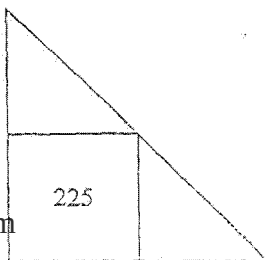
(Note: If the time is 8:45, the product of the digits is $(8)(4)(5) = 160$.)

- F. 90
- G. 162
- H. 405
- J. 708
- K. 729

19. The difference of 2 integers is 4. The sum of the same 2 integers is 38. What is the greater of the 2 integers?

- A. 17
- B. 18
- C. 19
- D. 20
- E. 21

20. In the figure below, the small isosceles right triangles are congruent. The area of the square is 225 square inches. What is the total area, in square inches, of the 2 small triangles?

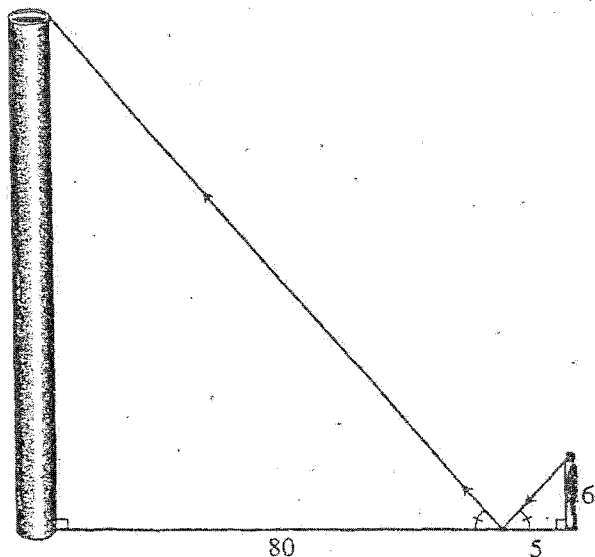


- F. 75
- G. 112.5
- H. 150
- J. www.crackab.com
- K. 450

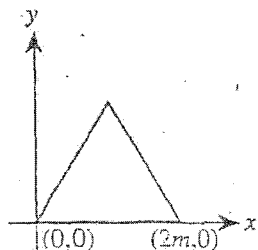


21. To measure the height of a smokestack, Ariel places a mirror on level ground 80 feet from the base of the smokestack. He then forms 2 similar triangles by looking into the mirror and backing up until he sees the reflection of the top of the smokestack, as shown below. Ariel's eyes are 6 feet from the ground and he is 5 feet from the image in the mirror. Which of the following is closest to the height, in feet, of the smokestack?

DO YOUR FIGURING HERE.



- A. 110
B. 96
C. 91
D. 85
E. 67
22. For what value of x , if any, is the equation $(x+5)^2 = (x+1)^2$ true?
F. -3
G. -1
H. 0
J. 3
K. There is no value of x for which the equation is true.
23. An equilateral triangle, with the coordinates of 2 vertices labeled, is shown in the standard (x,y) coordinate plane below. Which of the following are the coordinates of the 3rd vertex?



- A. $(0, m)$
B. (m, m)
C. $(m, m\sqrt{3})$
D. $(m\sqrt{3}, m)$
E. $(m\sqrt{3}, m\sqrt{3})$



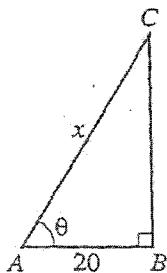
24. The diagonal of a square is $2\sqrt{2}$ inches long. What is the area of the square, in square inches?

DO YOUR FIGURING HERE.

- F. $\sqrt{2}$
 G. 2
 H. $2\sqrt{2}$
 J. 4
 K. $4\sqrt{2}$

25. In the figure below, $\overline{AC}$ represents a support wire x feet long. The wire is attached to a tower at C and to the ground at A . The distance between A and B , which is at the base of the tower, is 20 feet. The angle at A has measure θ . Which of the following relationships must be true?

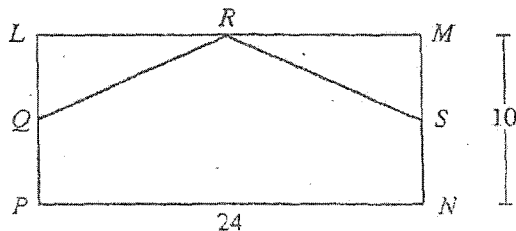
- A. $\cos \theta = \frac{20}{x}$
 B. $\sin \theta = \frac{20}{x}$
 C. $\sin \theta = \frac{x}{20}$
 D. $\tan \theta = \frac{20}{x}$
 E. $\tan \theta = \frac{x}{20}$



26. The equation $\sqrt{x} + \sqrt{x-16} = 8$ is true for what real value of x ?

- F. 9
 G. 16
 H. 25
 J. 36
 K. 64

27. In rectangle $LMNP$ below, $\overline{NP}$ is 24 inches long and $\overline{MN}$ is 10 inches long. Points Q , R , and S are the midpoints of $\overline{LP}$, $\overline{LM}$, and $\overline{MN}$, respectively. What is the perimeter, in inches, of pentagon $NPQRS$?



- A. 60
 B. 68
 C. 70
 D. 120
 E. 180

Use the following information to answer questions 28–30.

DO YOUR FIGURING HERE.

The table below gives data about the population of the U.S. for 1990, 1995, and 1999.

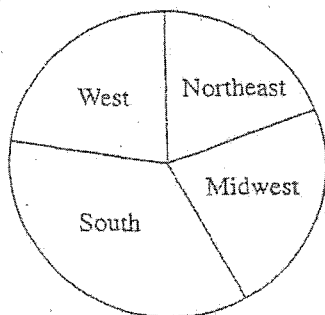
U.S. population category	1990	1995	1999
Group*			
Total	248.8	262.8	272.9
Male	121.3	128.3	133.4
Female	127.5	134.5	139.5
Age group†			
Under 18 years	25.7%	26.1%	25.7%
65 years and over	12.5%	12.8%	12.7%
Region‡			
Northeast	20.4%	19.6%	19.0%
Midwest	24.0%	23.6%	23.2%
South	34.4%	34.9%	35.4%
West	21.2%	21.9%	22.4%
*in millions of people, rounded to the nearest tenth of a million			
†in percent, rounded to the nearest tenth of a percent			

28. To the nearest percent, what percent of the total U.S. population in 1999 was male?

F. 50%
G. 49%
H. 48%
J. 47%
K. 20%

29. The circle graph (pie chart) below represents the 1999 U.S. population distribution for the 4 regions listed. To the nearest degree, what is the measure of the central angle of the "West" sector?

1999 Population Distribution



A. 81°
B. 76°
C. 68°
D. 40°
E. 22°

30. Expressed in millions of people, what was the average growth per year in total U.S. population from 1990 to 1995, rounded to the nearest 0.1 million?

F. 2.8
G. 2.7
H. 2.0
J. 1.4
K. 1.0

2



2

31. Every camera lens has a measurement called the focal length, f , such that when an object is in focus, the distance from the object to the center of the lens, D_o , and the distance from the center of the lens to the film, D_i , satisfy the equation $\frac{1}{D_o} + \frac{1}{D_i} = \frac{1}{f}$. If the object is in focus, $D_o = 36$ centimeters, and $D_i = 12$ centimeters, what is the focal length of the lens, in centimeters?

A. 3
B. 6
C. 9
D. 12
E. 24

DO YOUR FIGURING HERE.

32. What is the probability of randomly choosing a prime number from $\{2, 3, 4, 5, 6\}$?

F. 0
G. $\frac{1}{5}$
H. $\frac{2}{5}$
J. $\frac{3}{5}$
K. $\frac{4}{5}$

33. For $g(x,y) = 5x + 4y$, what is the value of $g(x,y)$ when $y = \left(\frac{2}{x}\right)^2$ and $x = 2$?

A. $\frac{23}{2}$
B. $\frac{49}{4}$
C. 16
D. 19
E. 28

34. What is the length, in coordinate units, of a diagonal of the rectangle that has $(0,3)$, $(4,3)$, and $(4,0)$ as 3 of its vertices in the standard (x,y) coordinate plane?

F. 3
G. 4
H. 5
J. 6
K. 7

35. What is the value of x if $\log_5 x = 2$?

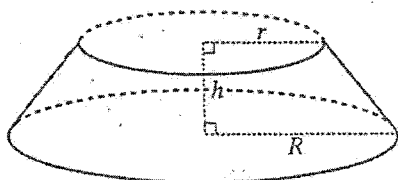
A. $\sqrt{5}$
B. $\sqrt[5]{2}$
C. 10
D. 25
E. 100



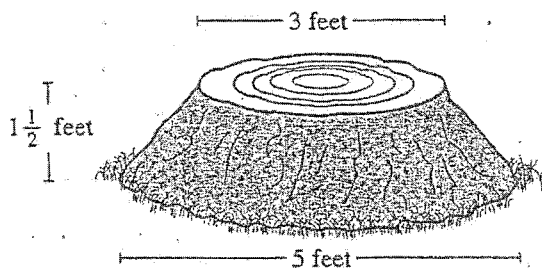
DO YOUR FIGURING HERE.

36. The fuel remaining in the gas tank of a particular car that starts with a full 16-gallon tank of gas can be approximated by the equation $y = 16 - 0.05x$, where x is the number of miles the car has traveled for $0 \leq x \leq 320$, and y is the number of gallons remaining in the tank. According to this equation, which of the following statements is true about this car?
- F. In traveling 0.05 miles, the car uses 1 gallon of gas.
 - G. In traveling 1 mile, the car uses 0.05 gallons of gas.
 - H. In traveling 16 miles, the car uses 0.05 gallons of gas.
 - J. In traveling 16 miles, the car uses 1 gallon of gas.
 - K. In traveling 80 miles, the car uses 16 gallons of gas.

37. A formula for the volume of a right circular cone with its top removed is $V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$, where R and r are radii and h is height, as shown in the figure below.



The number of cubic feet of wood in a tree stump can be estimated using the formula given. About how many cubic feet of wood are there in the tree stump shown below?



- A. 12
 - B. 18
 - C. 19
 - D. 38
 - E. 77
38. Which of the following is the set of real solutions for the equation $6x + 10 = 2(3x + 5)$?
- F. The empty set
 - G. $\{-\frac{5}{3}\}$
 - H. $\{0\}$
 - J. $[0, 1]$
 - K. The set of all real numbers

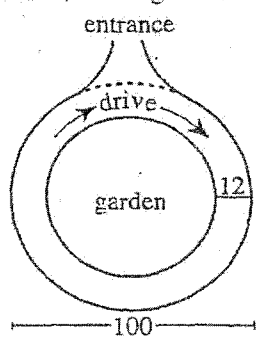


DO YOUR FIGURING HERE.

39. The expression $\frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{3}}$ equals:

- A. $\frac{36}{65}$
- B. $\frac{13}{20}$
- C. $\frac{18}{13}$
- D. $\frac{20}{13}$
- E. $\frac{8}{3}$

40. The figure below shows a circular garden surrounded by a circular drive. The outside circle of the drive has a diameter of 100 meters and the drive has a uniform width of 12 meters. Which of the following is closest to the area, in square meters, of the garden?

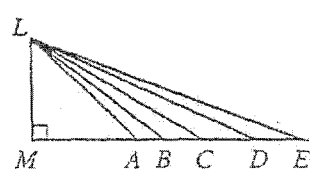


- F. 240
- G. 310
- H. 4,500
- J. 7,900
- K. 18,000

41. For all nonzero real numbers x and y , what is the value of $x^0 + y^0$?

- A. 2
- B. 1
- C. 0
- D. $x + y$
- E. Undefined

42. In the figure below, $LM = MA$, and A , B , C , and D are on $\overline{ME}$. Of the angles $\angle LAM$, $\angle LBM$, $\angle LCM$, $\angle LDM$, and $\angle LEM$, which one has the smallest sine?



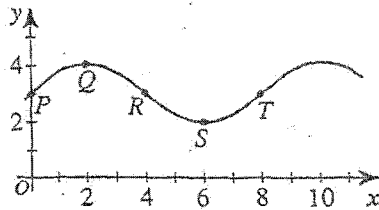
- F. $\angle LAM$
- G. $\angle LBM$
- H. $\angle LCM$
- J. $\angle LDM$
- K. $\angle LEM$



Use the following information to answer questions 43–45.

DO YOUR FIGURING HERE.

The graph of $y = f(x)$ is shown in the standard (x, y) coordinate plane below with 5 points labeled.



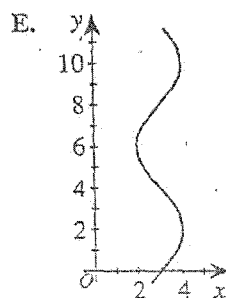
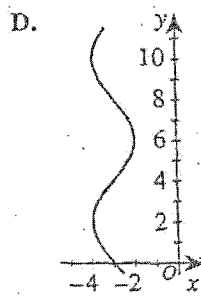
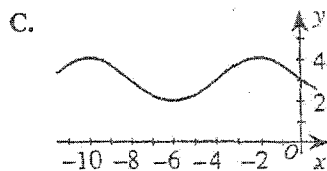
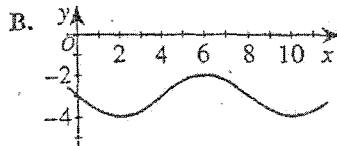
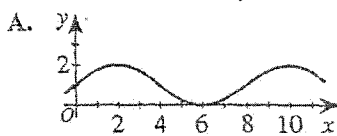
43. Which of the following points is the y -intercept of the graph of $y = f(x)$?

A. P
 B. Q
 C. R
 D. S
 E. T

44. The function $y = f(x)$ can be classified as one of the following types of functions. Which one is it?

F. Constant
 G. Linear
 H. Quadratic
 J. Cubic
 K. Trigonometric

45. Which of the following graphs is the reflection of $y = f(x)$ across the line $y = x$?





46. If x is a factor of 35 and y is a factor of 16, the product of x and y could NOT be which of the following?

DO YOUR FIGURING HERE.

- F. 1
- G. 24
- H. 40
- J. 112
- K. 560

47. For each positive integer n , let n_e be the product of all positive even numbers less than or equal to n . For example, $6_e = (6)(4)(2) = 48$ and $7_e = (6)(4)(2) = 48$.

What is the value of $\frac{12_e}{3_e}$?

- A. 384
- B. 1,280
- C. 1,920
- D. 15,360
- E. 23,040

48. If $(1, b)$ is on the graph of the equation $x - 3y = 7$ in the standard (x, y) coordinate plane, then $b =$?

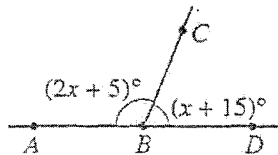
- F. $-\frac{8}{3}$
- G. -2
- H. 2
- J. 3
- K. 6

49. For all $x > 0$, what happens to the value of $\frac{x^2 + x}{x} - x$, as x increases?

- A. It decreases in proportion to x^2 .
- B. It decreases in proportion to x .
- C. It increases in proportion to x .
- D. It increases in proportion to x^2 .
- E. It remains constant.

50. In the figure below, B is on $\overline{AD}$ and the angle measures are as given. What is the degree measure of $\angle ABC$?

- F. $111\frac{2}{3}^\circ$
- G. $106\frac{2}{3}^\circ$
- H. $68\frac{1}{3}^\circ$
- J. $53\frac{1}{3}^\circ$



- K. 25°



DO YOUR FIGURING HERE.

51. In the standard (x,y) coordinate plane, what is the slope of a line passing through the points $(-1,-8)$ and $(0,6)$?

A. -14
 B. -2
 C. $\frac{1}{14}$
 D. 2
 E. 14

52. What is the degree measure of an angle that measures

$\frac{5\pi}{12}$ radians?

F. 15°
 G. 75°
 H. 150°
 J. $\left(180 - \frac{5\pi}{12}\right)^\circ$
 K. $\left(\frac{360 - 5\pi}{12}\right)^\circ$

53. A circle in the standard (x,y) coordinate plane has its center at $(2,6)$ and an area of 4π square coordinate units. Which of the following is an equation for this circle?

A. $(x-6)^2 - (y-2)^2 = 16$
 B. $(x-6)^2 + (y-2)^2 = 4$
 C. $(x-6)^2 + (y-2)^2 = 16$
 D. $(x-2)^2 + (y-6)^2 = 4$
 E. $(x-2)^2 + (y-6)^2 = 16$

54. Which of the following is FALSE for some x and y that satisfy the equation $x\left(\frac{1}{y}\right) = 1$?

F. $y\left(\frac{1}{x}\right) = 1$
 G. $x = y$
 H. $x^2 + y^2 = 2xy$
 J. $x^2 = y^2$
 K. $x^5 + y^5 = 1$



55. The 2×3 matrix $\begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & -2 \end{bmatrix}$ represents $\triangle ABC$ with vertices $A(-2,1)$, $B(1,2)$, and $C(3,-2)$ in the standard (x,y) coordinate plane. After a translation of $\triangle ABC$, the matrix representing the translated triangle is $\begin{bmatrix} -4 & -1 & 1 \\ 0 & 1 & k \end{bmatrix}$. What is the value of k ?

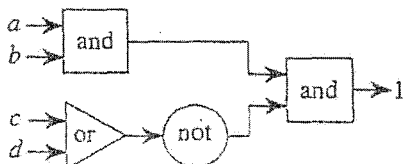
A. -1
B. -2
C. -3
D. -4
E. -5

DO YOUR FIGURING HERE.

56. Three different logical operators are defined in the table below.

Symbol	Operator	Description
	AND	If both inputs are 1, then the output is 1. Otherwise, the output is 0.
	OR	If either input is 1 or if both inputs are 1, then the output is 1. Otherwise, the output is 0.
	NOT	If the input is 0, then the output is 1. Otherwise, the output is 0.

The logic diagram below uses the 3 operators. The only possible values for a , b , c , and d are 0 and 1. Which of the following (a,b,c,d) inputs will result in an output of 1?

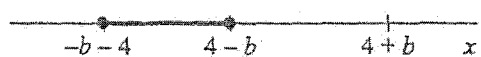
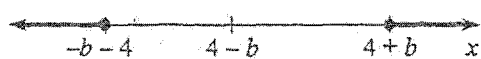
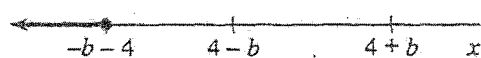
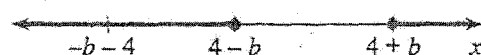


F. (0,1,0,0)
G. (0,1,1,1)
H. (1,0,1,0)
J. (1,1,0,0)
K. (1,1,0,1)



57. Whenever $b > 0$, which of the following real number line graphs represents the solutions for x to the inequality $|x + b| \geq 4$?

DO YOUR FIGURING HERE.

- A. 
- B. 
- C. 
- D. 
- E. 

58. Whenever m and n are both integers, what is $(5.0 \times 10^m)(8.0 \times 10^n)$ expressed in scientific notation?

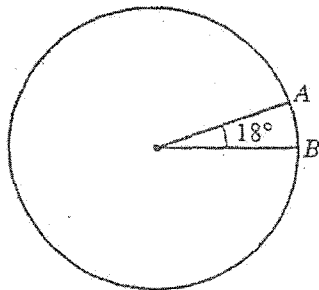
- F. 4.0×10^{mn}
 G. $4.0 \times 10^{m+n+1}$
 H. 40.0×10^{mn}
 J. $40.0 \times 10^{m^n}$
 K. 40.0×100^{mn}

59. The points A , B , C , and D lie in that order on a straight line. The midpoint of $\overline{AC}$ is B , and the midpoint of $\overline{AD}$ is C . The length of $\overline{AB}$ is x inches, and the length of $\overline{CD}$ is $3x - 6$ inches. What is the length, in inches, of $\overline{AD}$?

- A. 24
 B. 16
 C. 12
 D. 8
 E. 6

60. The circle shown below has a radius of 10 cm. A central angle with measure 18° intercepts minor arc $\overline{AB}$. How many centimeters long is minor arc $\overline{AB}$?

- F. $\frac{1}{10}\pi$
 G. $\frac{1}{5}\pi$
 H. π
 J. $\frac{100}{18}\pi$
 K. 180π



END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are four passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from the short story "My Father's Son" by Steven Corbin (©1996 by The Estate of Steven Corbin), which appeared in the anthology *Streetlights: Illuminating Tales of the Urban Black Experience*.

When I pick up my father and his girlfriend at Los Angeles International Airport he stops short before embracing me, blinks repeatedly in disbelief, smiles or smirks ~~www.ck12.com~~ sardonically, judging me, sizing me up through his trademark horn-rimmed glasses and ultraconservative mud-brown eyes, which cut immediately to the wiry dreadlocks bouncing around my forehead, and his glare that screams *What kinda weird thing you gone and done to yourself this time?* disturbingly familiar yet intimidating and benevolent at the same time; reduces me, as it most often does despite my thirtysomething years, to the little tyke in dungarees and high-top sneakers I once was, in dire need of my father's love and approval. For weeks I was preoccupied with and vexed myself over what my father and I would possibly talk about for seven days and seven nights, constantly being together and sharing each meal, sleeping in the same house day in, day out, after our years of estrangement sealed by an impromptu reconciliation. But now that he's here and we move toward one another, I suddenly acknowledge that seven days might not be sufficient time to encompass all that we have to say to each other.

His girlfriend Delores scoops me up in her arms, delighted to see me again. My father stands off to the side, nervously awaiting his turn, large, rugged, callused hands thrust deep inside his pants pockets, as he psyches himself up for a fleeting, ephemeral moment of intimacy with another adult male. Surprising me, he locks me in an embrace that is both macho and compassionate and he doesn't let go first, which is foreign to me since I can count, on one hand, the paucity of occasions when my father has embraced me; and in his touch I can feel, almost palpably, his gratification in seeing his firstborn for the first time since two Christmases ago. Before leaving the terminal, Delores insists upon taking a photo of us, father and son, day one of the arrival, my arm wrapped around him and the rigidity of his pose, the tightening of his muscles and the tautness of his spine, that won't be captured by the camera's aperture and recorded on the celluloid.

As I leave the airport terminal, my father and Delores exchange gut-level reactions and first impressions of Los Angeles, my father in the front seat, Delores in the back. My father leans over and switches on the radio, predictably, frantically searching the dial for the local jazz station. His horrified expression freezes as I explain, in the apologetic tone of a parent informing the child that the ice-cream vendor won't be coming today, that Los Angeles' only jazz station is now defunct. But I believe, I say quickly, in an effort to smooth out the wrinkles in his face, that there's another noncommercial broadcast from one of the local universities. His radar ever intact, he finds the broadcast. His fingers stop moving only when the dial slides into Miles Davis's muted horn blaring out at us. The disc jockey is spinning a recent composition. My father and I inhale deeply, both of us immediately recognizing the signature horn, pretending we don't, my foot and fingers tapping to the rhythm of genius, as we consciously sidestep the potential argument we engage in from time to time regarding the merit of Miles's contemporary contributions. My father, jazz elitist and purist, firmly and unequivocally and inexorably theorizes that the legendary horn blower has done absolutely nothing musically worthwhile recently. He sold out, my father upholds. He's growing, I contradict. We can't and shouldn't expect him to repeat what he did in the fifties and sixties.

Studying my father at red lights and through intermittent glances that punctuate our conversation, I realize, understand, and accept, as I never have before, that he is indeed aging. Still remarkably handsome, even dashing for his middle age, his silky hair, short and black and wavy and parted at the side as it was the day he married my mother at age twenty, is his sole physical attribute that remains timeless. When I became a teenager, then segued into young adulthood, the maturation of my growth having kicked in, shaping and molding the adult face through which I would see the world, a face that would pretty much stay the same forever and always, people began physically comparing me to my father, a comparison it took me awhile to get used to. Even my mother would look at me, shake her head, smile as she *hum-hummed* to herself at the alleged stark resemblance I could never see. You are your father's son all right, she'd say.

1. The central concern presented in the passage is:
 - A. the acknowledgment by two very different people that they will never get along.
 - B. the reunion and reconciliation between a father and a son.
 - C. a father's first impression of the city where his son has chosen to live.
 - D. the struggle of a father to overcome the misconceptions he holds about his son.
2. As revealed in the passage, the father is best described as:
 - F. restrained and unforgiving.
 - G. opinionated and flamboyant.
 - H. selfless but impulsive.
 - J. intimidating but compassionate.
3. The author implies the emotional state of the characters primarily through:
 - A. the use of extensive dialogue.
 - B. an objective and omniscient narrator.
 - C. physical descriptions and narrative reflection.
 - D. detailed flashbacks and elaborate settings.
4. The point of view from which this passage is told can best be described as that of a:
 - F. son who as an adult narrates events as they unfold.
 - G. son who as a child recalls his authoritative father.
 - H. father who nervously anticipates a reunion with his son.
 - J. father and his girlfriend who travel to visit his young son.
5. It can reasonably be inferred from the second paragraph (lines 24–41) that the narrator is surprised by:
 - A. his father's nervousness at their meeting.
 - B. his father's desire not to be photographed.
 - C. Delores's role as a neutral mediator.
 - D. the deep feeling transmitted in his father's embrace.
6. The interactions between the narrator's father and the narrator in lines 36–41 reveal a relationship that is best described as one that:
 - F. grows closer daily, now that the father has moved to Los Angeles.
 - G. seems to be strained despite the efforts of the father and son to reunite.
 - H. has been restored to its former affectionate state that both remember from the past.
 - J. survives despite the son's desire to distance himself from his father.
7. The main purpose of the first paragraph is to provide:
 - A. a description of the long-standing personal dynamic between the narrator and his father.
 - B. background information about the father's life previous to this meeting.
 - C. an explanation of the recent events that prompted the father's trip.
 - D. clarification about which character planned and initiated this family visit.
8. As revealed in the first paragraph, in this meeting with his father, the narrator realizes for the first time which of the following?
 - F. He needs his father's love and approval.
 - G. His father's facial expressions can make the narrator feel like a child again.
 - H. His father does not approve of the narrator's personal style choices.
 - J. He has more to share with his father than time might allow.
9. As it is used in line 31, the word *foreign* most nearly means:
 - A. unrelated.
 - B. incompatible.
 - C. uncharacteristic.
 - D. far away.
10. The narrator considers his father's assessments of the newer jazz works of Miles Davis to be:
 - F. strong and hasty.
 - G. old-fashioned and held by most listeners.
 - H. predictable and in opposition to his own.
 - J. convincing and forceful.

Passage II

SOCIAL SCIENCE: This passage is adapted from *Speaking into the Air: A History of the Idea of Communication* by John Durham Peters (©1999 by the University of Chicago).

Strangely enough, little research in media history has been done on correspondence by letter. Media historians are beginning to take the post office seriously as a key site for understanding the development of communications. The cultural history of the mails is a remarkably rich source for philosophical visions of the varieties of communicative experience.

The notion that the mails involve delivery of a private, specifically addressed message was late in evolving. The current division of genres between personal and public correspondence did not exactly exist in the eighteenth-century newsletter in England and the colonies. The "familiar" letter was distinct from the newsletter, the forerunner of the modern newspaper, but both could be edited for and by the public. Newsletters had very high pass-along rates; they were meant quite literally to circulate among readers who would hand-write additional notices in blank spaces left for that purpose. In a similar way, personal letters in the United States could be raided for publication in the newspaper or at least for postmaster-led discussion. Some postmasters in the colonial period apparently freely quoted in their newspapers from love letters and personal correspondence. Not only was content open to stray eyes, but the receipt of mail was itself public because local post offices in the United States routinely kept log-books on who purchased postage for what mail, since payment was typically made by the recipient rather than the sender before the 1850s. Hence not only were local postmasters well informed on local reading habits, they were privy to much of the news locally in circulation and often monitors, even censors, of what newspapers local postal patrons would read and what mail they would receive. The post was not a secure channel. Letters then were more like postcards today—both privately addressed and publicly accessible.

Since the mid-nineteenth century, postal practices in North America and Western Europe quite explicitly sought to contain the potential for straying missives by giving senders private control over their letters and making the address circuitry much more focused. The key innovations that took place in the two middle decades of the century made the modern private letter possible. The first postage stamp appeared in 1840 in Great Britain, bearing a portrait of Queen Victoria. No longer did one need to see a postmaster to pay for carriage, marking a key step toward impersonality in access. In the 1840s adhesive postage stamps appeared in the United States, first as local, private issues, and in 1847 the first national stamp was authorized by the United States Congress. The first United States patent for envelopes was issued in 1849. By sealing off contents against inspection, envelopes gave letters an entirely new aura of privacy. In 1851 Congress, perhaps motivated to secure linkage with the Pacific Coast in

the wake of the 1849 gold rush, passed a flat rate for all letters, not graded for distance as some early rates were. In 1856 all mail in the United States had to be prepaid (as opposed to COD, or cash on delivery), and a registered mail service was founded to help prevent the loss of valuables, though it was rarely used. In 1858 street drop boxes, introduced in London in 1855, were first used in the United States. By the late 1850s, then, it was possible to mail a letter sealed in an envelope, paid for with a prepurchased stamp, and dropped into a public box. "No longer did the sender have to come under the scrutiny of the receiving postal employees" (Mathew J. Bowyer). No sentinels guarded the gates to the system. Confidentiality was now possible.

Here, then, we have a system of public communication, connected to every address in the nation, that allows for the conveyance of private messages in sealed packages. Mail, the circulation system of writing and other lightweight cargo, was no longer locally inspected to the same degree. Stamps, envelopes, and drop boxes made the individual sender in principle sovereign over the letter. The post office had thus made an inherently public and plural signifying system into one governed by the private will of the sender. The post office, by accommodating sender-imposed restrictions on receivers, had transformed letters from creatures of dissemination into creatures of apparent dialogue.

11. The main idea of the passage is that:

- A. the mail system of the colonial United States gave no privacy to letter senders and even allowed their letters to be printed in newspapers.
- B. it was only after envelopes were introduced that letter senders in the United States were assured their mail was confidential.
- C. the history of the mail system in the United States is highlighted by the evolution of letters from being public to private.
- D. research into the history of the mail system in the United States shows that the transition of letters from being public to private was prompted by innovations in Great Britain.

12. The main idea of the third paragraph (lines 37–69) is that:
- F. mid-nineteenth-century innovations giving senders more power over the contents and handling of letters made the private letter of today possible.
 - G. the introduction of prepaid mail service in the mid-nineteenth century greatly increased the popularity and convenience of the U.S. postal system.
 - H. the increased confidentiality of the personal letter in the mid-nineteenth century made censorship of letters by postmasters unlikely.
 - J. because letters sent after the mid-nineteenth century no longer came under the scrutiny of postmasters, letters began to carry different information.
13. As it is used in lines 81–82 to describe letters, the term *creatures of dissemination* most nearly means that letters:
- A. allowed two people to have a written conversation with each other.
 - B. were the vehicles used by postmasters to censor information.
 - C. provided information to many people.
 - D. tended to start rumors.
14. According to the passage, which of the following was the final innovation that allowed letters to be sent without postmaster scrutiny?
- F. Street drop boxes
 - G. Registered mail
 - H. Sealed envelopes
 - J. Postage stamps
15. According to the passage, which of the following statements is true of colonial newspapers?
- A. They sometimes contained quoted material from personal letters.
 - B. They were usually purchased at the post office.
 - C. They were generated by postmaster-led discussions.
 - D. They often printed misquotes and untruths.
16. Information in the second paragraph (lines 8–36) makes it clear that postmasters in colonial times:
- F. were volunteers and therefore did not get paid.
 - G. had duties very similar to those of postmasters today.
 - H. were often newspaper editors as well.
 - J. opposed the idea of adhesive postage stamps.
17. The passage most strongly suggests that prior to the 1850s the recipient of a letter would:
- A. have been quite certain the postmaster would have censored the letter.
 - B. likely have had to open an envelope to read the letter.
 - C. have been unfamiliar with the sender of the letter.
 - D. likely have had to pay for receipt of the letter.
18. According to the passage, when was a U.S. patent issued for envelopes?
- F. Prior to the introduction of the first postage stamp in Great Britain
 - G. After the introduction of adhesive postage stamps in the United States
 - H. In the same year registered mail was introduced in the United States
 - J. After the introduction of street drop boxes in London
19. The passage states that the same year registered mail was introduced in the United States, all mail had to be:
- A. inspected by a postmaster.
 - B. collected from a drop box.
 - C. contained in an envelope.
 - D. paid for in advance.
20. According to the passage, registered mail service in the United States was:
- F. prompted by the 1849 gold rush.
 - G. made possible by the introduction of prepaid postage stamps.
 - H. based on an earlier London model.
 - J. rarely used.

Passage III

HUMANITIES: This passage is adapted from the essay "Erased Edges" by Barbara Hurd (©2001 by Barbara Hurd).

Once, I enrolled in a weeklong seminar to learn how the government delineates wetlands. The government's guide to finding the edge of a swamp is fifty pages long, complete with graphs and soil maps you need a magnifying glass to decipher. But a liquid landscape cannot be nailed down with maps and charts any more than love can be understood as a biochemical reaction of pheromones.

Trying to define the edge of a swamp is like trying to put a neatly folded shadow into a dresser drawer. Our efforts to outline these places arise from a desire for tidiness, a wish that nothing undefined lurk around our own edges. But the truth is, even human boundaries shift. Some mornings, I wake up feeling small and compact. I barrel through those days, teaching, opening mail, and grading papers like a self-propelled lawn mower. Other days, I feel huge, airy, globular. Someone blows through the bubble wand, and I billow out my front door, roll into the garden, into class. I can't grade a thing, but I can see the shadow a poem makes, where it wants to go. On those days, I can live with how the edges of a swamp shift, how its underbelly can sometimes surge in the center and open up into a pond, how the pond can disappear and the woods around the perimeter go soggy overnight.

Once, as children, my twin sister and I fashioned our own seedpod by threading an old rope through the four corner grommets of a tarpaulin and flinging the rope's frayed end over a large limb of a maple in front of the house. We sat cross-legged in the middle of the tarp and pulled on the rope until the four corners rose around us, enclosing us; and the sack began to rise off the ground. We were actually able to hoist ourselves a few feet into the air this way and dangle there, a khaki-green sack of twins, a pod hanging heavy on the vine, its seeds hidden inside, suspended and swaying over the front lawn of a suburban neighborhood.

At that time, my family lived just a half mile or so from a small pond. One spring I spent weeks there imagining my stick was a giant pencil with a pink rubber eraser on the end. I imagined crouching on the bank of the pond and systematically erasing the edge. There was something quite magical about the notion that you could change the ground you stood on by erasing some line.

The gift of ambiguity is that it stretches us. Makes us less rigid. Nudges us out of either/or thinking. One morning when I was forty-two, I stood in my bedroom, changing from a sweat shirt to a T-shirt, and suddenly I couldn't remember whether my twin or I was older. Was I Baby Girl A or Baby Girl B? From the moment of our birth, in spite of our parents' best efforts to differentiate us, we were "the twins," a unit who shared bedrooms and friends and chores and clothes and good-

night rituals, a unit others momentarily dissected by asking which of us was older, as if those thirteen minutes were the only distinction between us. So, on the midlife morning when I stood in my bedroom and felt those thirteen minutes utterly dissolve, the realization swept over me that if I didn't know which one of us was older, if that distinction disappeared, then I might very well be my sister, not me. Looking in the mirror didn't help. I recognized the face, knew this body was the writer's and that other one the designer's. But I felt somehow it was quite possible that I had momentarily sloshed into her body and she into mine, that this might not be me now sitting on the bathroom floor, the way if you take a stick and dig a channel between puddles in the dirt, the water from each puddle heads toward the other, and there's nothing to stop the exchange of liquid so that, in just an instant, there's some new, double-bulged body of water, but no recognizable puddle A or puddle B.

Here's what I love about the in-between: its inherent ambiguity, how it invites a swaying of the imagination, a languid hammock swing between two definitions, two identities. The mind, like the body, swings sideways, rises over one bed of possibilities, pauses at the peak, considers, sweeps down and over to the other side. I am one twin, then the other. This swamp is land first, and then water; this turtle in my hand a waddler, and now a swimmer. We are shape-shifters, all of us, liquid mosaics of mutable and transient urges, and we give ourselves headaches when we pretend otherwise, when we stiffen into permanent and separate identities unsullied by the drifting slop, the very real ambiguities of ourselves and the world.

21. In the passage, the author refers to herself as all of the following EXCEPT:

- A. her twin sister.
- B. a shape-shifter.
- C. a neatly folded shadow.
- D. a self-propelled lawn mower.

22. The author would most likely agree with which of the following statements about the seminar mentioned in the first paragraph?
- F. The organizers were poorly prepared to teach what most of the students expected to learn.
 - G. It provided a substantial amount of information about wetlands, but it fell short of satisfying the author's curiosity.
 - H. It was an exciting introduction to wetlands management and motivated the author to do graduate work in the field.
 - J. It successfully combined scientific information with accounts of personal experiences in wetlands.
23. As it is used in the passage, the phrase "a biochemical reaction of pheromones" (lines 7-8) represents:
- A. a government-sponsored wetland evaluation.
 - B. an exchange the narrator observes in swamps.
 - C. the focus of the weeklong seminar.
 - D. an absurdly insufficient definition of love.
24. The word *seeds* in line 36 refers to the:
- F. author's ideas.
 - G. author and her sister.
 - H. next generation of wetland grasses.
 - J. fragile universe.
25. The passage states that the author's twin sister is by profession a:
- A. designer.
 - B. biologist.
 - C. writer.
 - D. business owner.
26. Which of the following best describes how the author uses the reference to the turtle (line 81)?
- F. She offers the turtle as an image that represents humans' slow progress toward understanding.
 - G. She contrasts the turtle with humans who rush thoughtlessly through life.
 - H. She uses the turtle as another example of a living thing that has more than one identity.
 - J. She compares the turtle's shell to the rigid ideas that people use to protect themselves from uncertainty.
27. Information in the passage indicates that the author's age when she wrote this passage was:
- A. twenty-two or younger.
 - B. between twenty-two and thirty.
 - C. between thirty and forty.
 - D. forty-three or older.
28. Which of the following statements about the author and her twin sister as children is supported by the passage?
- F. They resented being mistaken for one another and attempted to form very distinct identities.
 - G. They spent a great deal of time together and shared possessions, activities, and friends.
 - H. The author continually imagined that she was herself one minute and her sister the next.
 - J. They were constantly reminding others that they were born thirteen minutes apart.
29. When the author uses "we" in line 84, she is most likely referring to:
- A. her twin sister and herself.
 - B. all twins.
 - C. everyone in her family.
 - D. people in general.
30. Which of the following would the author most likely see as an example of what it means to "pretend otherwise" as the phrase is used in line 85?
- F. A twin pretending to be his or her twin sibling
 - G. A biologist pretending to be a writer
 - H. A human pretending to be a turtle
 - J. A singer pretending to be nothing but a singer

Passage IV

NATURAL SCIENCE: This passage is adapted from the book *Biomimicry: Innovation Inspired by Nature* by Janine M. Benyus (©1997 by Janine M. Benyus).

It's a steamy 80 degrees Fahrenheit in Christopher Viney's lab, in deference to a six-inch-long golden orb weaver spider, who is dining on crickets while being silked. A gossamer thread issues from her enormous abdomen at a steady clip, wound by a motor onto a revolving spindle. In this session alone, she will donate about 100 feet of "dragline," a specialty silk designed for rappelling from drop-offs and framing the spokes and perimeter of her web.

Compared ounce to ounce with steel, dragline silk is five times stronger, and compared to Kevlar (found in bulletproof vests), it's much tougher—able to absorb five times the impact force without breaking. Besides being very strong and very tough, spider silk also manages to be highly elastic, a hat trick that is rare in any one material. If you suspend increasingly heavy weights from a steel wire and a silk fiber of the same diameter, their breaking point is about the same. But if a gale force wind blows, the strand of silk (five times lighter in weight) will do something the steel never could—it will stretch 40 percent longer than its original length and bounce back good as new. Up against our stretchiest nylon, spider silk bungees 30 percent farther.

Another characteristic in spider silk's favor is that it has to get very, very cold before it becomes brittle enough to break easily. In the frigid temperatures that parachutes encounter, for instance, spider silk would make ideal lightweight lines. Other uses would be cable for suspension bridges, artificial ligaments, and sutures. But how would we fit so much function into such a small package?

Spider silk begins as a pool of raw liquid protein sloshing around in a gland. Viney hypothesizes that the raw liquid silk leaves the gland and travels through a thin duct just before entering the spinneret. As it squeezes through the duct, water is wrung out of the protein and calcium is added. The globules hook up in a pop-bead necklace, making the solution one thousand times less viscous, because the rodlike assemblies can now slide past one another. It's analogous to putting lanes of traffic on a highway sliding past one another, versus the mess that is a laneless, lawless traffic jam.

Viney's model has a pleasing simplicity and completeness: The globular proteins line up into a pop-bead necklace, which squeezes through the spinneret to become a silk fiber. The final product is partly flexible and partly rigid, like a reinforced Slinky. The amorphous part gives, but the stiff crystalline domains don't give. When the fiber becomes notched, a crack or tear gets interrupted by the crystalline regions and can't propagate.

Some, like silk researcher Randy Lewis, don't agree with Viney's model. Lewis feels he has evidence

that there are actually two proteins rather than just one that make up spider silk. "In the two-protein hypothesis, Viney's pop-bead model doesn't make sense," says Lewis. But other researchers are still not convinced of the existence of two proteins. While the jury is out and the debate is lively, all the investigators in spider silk research encourage one another to keep theorizing. When you think about what it could mean in terms of sustainable fiber manufacture, this research, tough as it is, is definitely worth it.

Consider: The only thing we have that comes close to silk in quality is polyaramid Kevlar, a fiber so tough it can stop bullets. But to make Kevlar, we pour petroleum-derived molecules into a pressurized vat of concentrated sulfuric acid and boil it at several hundred degrees Fahrenheit in order to force it into a liquid crystal form. We then subject it to high pressures to force the fibers into alignment as we draw them out. The energy input is extreme, the toxic by-products odious.

The spider manages to make an equally strong and much tougher fiber at body temperature, without high pressures, heat, or corrosive acids. Best of all, says Viney, spiders don't have to drill offshore for oil to produce the silk. They take flies and crickets at one end and process a high-tech material at the other end.

If we could learn to do what the spider does, we could take a soluble raw material that is infinitely renewable and make a super-strong water-insoluble fiber with negligible energy inputs and no toxic outputs. We could apply that processing strategy to any number of fiber precursors. Imagine what it would do to our fiber industry, which is now heavily dependent on petroleum, both for raw material and processing! To break that dependency, says Viney, we have to become spider's apprentices.

31. Which of the following questions is NOT answered by the passage?

- A. Why is the pop-bead model unworkable if there are two proteins in spider silk?
- B. Does the author have an opinion as to the value of the research done by Viney and Lewis and others like them?
- C. What do spiders eat?
- D. Can a spider produce silk at body temperature?

32. It is most reasonable to infer that if the apprentices referred to in line 89 were successful, which of the following would occur?
- F. Viney would become a research apprentice to Lewis.
 - G. The amount of toxic waste produced by the fiber industry would decrease.
 - H. Liquid protein would become more available at supermarkets.
 - J. The research into spider silk would come to a halt in all but a few obscure labs.
33. Which of the following best describes how the phrase "laneless, lawless traffic jam" (line 42) functions in the passage?
- A. As an example of what a spider web looks like to the untrained eye
 - B. As an analogy for the heated disagreement between scientists about the process of spider silk manufacture
 - C. As an image for the structure of spider silk were the silk not formed into rodlike assemblies
 - D. As a suggestion of how Viney's office might appear to an outsider
34. Which of the following statements best describes Kevlar and its significance in relation to spider silk, according to the passage?
- F. Spider silk could replace Kevlar, the tough material in bulletproof vests, but the production process would yield many odious by-products.
 - G. Kevlar, the material that makes bulletproof vests possible, has all of the strength but none of the flexibility of spider silk.
 - H. Of human-made materials, Kevlar comes closest in quality to spider silk, but its manufacture takes a heavy toll on natural resources.
 - J. Kevlar, a material used in bulletproof vests, was what inspired Viney to focus his research on the structure and function of spider silk.
35. As it is used in line 4, the word *issues* most nearly means:
- A. takes.
 - B. separates.
 - C. stitches.
 - D. emerges.
36. The passage states that which of the following combinations of qualities rarely occurs in the same material?
- F. Extreme strength and extreme elasticity
 - G. Extreme lightness (in weight) and extreme strength
 - H. Extreme water resistance and extreme elasticity
 - J. Extreme viscosity and extreme durability
37. What comparison does the author make between spider silk and nylon?
- A. When spider silk is stretched out, its molecular structure resembles that of nylon.
 - B. Spider silk is one-third as stretchy as the most stretchy nylon.
 - C. Spider silk is as stretchy as the most stretchy nylon.
 - D. Spider silk is 30 percent more stretchy than the most stretchy nylon.
38. According to Viney's model as it is presented in the passage, what is the route that molecules travel on their way to becoming spider silk?
- F. Gland, duct, spinneret
 - G. Spinneret, gland, duct
 - H. Gland, spinneret, duct
 - J. Duct, gland, spinneret
39. As it is used in line 78, the word *take* most nearly means:
- A. steal.
 - B. acquire.
 - C. consume.
 - D. choose.
40. According to the passage, the fiber industry is heavily dependent on:
- F. silk.
 - G. petroleum.
 - H. government subsidies.
 - J. rodlike assemblies.

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.

SCIENCE TEST
35 Minutes—40 Questions

DIRECTIONS: There are seven passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage I

Scientists examined a continuous 40,000-year fossil record of a snail species that lives today on Bonin Island in the Pacific Ocean. Fossil snail shells, along with living snails' shells, were studied to determine how the snails have changed over time.

Twenty complete fossil snail shells were randomly extracted from each rock layer on the island. The shells were cleaned, photographed, and measured. A small piece of each shell was used to determine the shell's age by radiocarbon dating.

Table 1			
Age of shells (years before present)	Percent of shells with the following pattern of stripes		
	L, L, L, B	L, L, B, B	B, L, B, B
40,000	33	20	47
33,000	86	6	8
32,000	79	9	12
30,000	72	13	15
26,000	69	15	16
24,000	50	0	50
22,000	0	0	100
18,000	0	0	100
16,000	0	0	100
12,000	0	0	100
6,000	0	0	100
1,500	21	5	74
900	13	7	80
0	4	8	88

Study 2

The number of *whorls* (turns of the shell as viewed from above), the inner lip length, and the diameter of the shell were measured for each shell (see Figure 1).

Study 1

Living snails had dark-colored stripes on their light-colored shells. The stripes on the fossil shells were not visible to the naked eye, but could be seen when the shells were viewed under ultraviolet light. The pattern of stripes on each shell was recorded, starting at the top of the shell.

Any stripe having a width less than $\frac{1}{10}$ the height of the shell was called a *line* (L) and any stripe exactly $\frac{1}{10}$ the height of the shell or wider was called a *band* (B). The ages of the shells and the percent of the shells of each age with a given pattern of stripes are shown in Table 1.

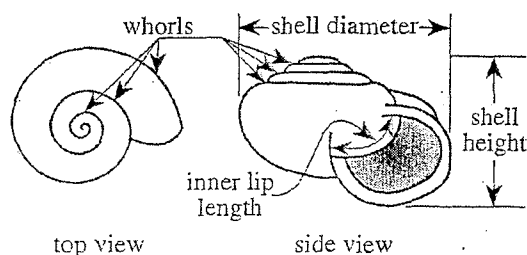


Figure 1

The average of each of the 3 geometric characteristics for shells of a given age was calculated. These results are shown in Figure 2.

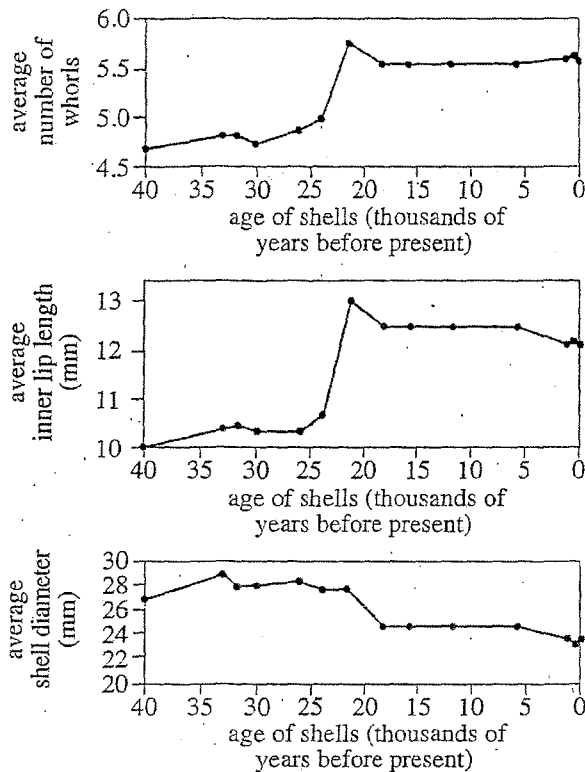


Figure 2

Table and figures adapted from Satoshi Chiba, "A 40,000-year Record of Discontinuous Evolution of Island Snails." ©1996 by The Paleontological Society.

- According to the results of Study 2, the average shell diameter of fossil snail shells with an age of 30,000 years is closest to which of the following?
 - 24 mm
 - 26 mm
 - 28 mm
 - 30 mm
- Suppose, in Study 1, the scientists had determined that the age of fossil snail shells taken from a particular rock layer on the island was 31,000 years. Based on the results of Study 1, the stripe pattern percents for that group of shells would most likely have been closest to which of the following?

	L, L, L, B	L, L, B, B	B, L, B, B
F.	100%	0%	0%
G.	75%	11%	14%
H.	65%	15%	20%
J.	50%	25%	25%

- Which of the following statements best describes how Study 1 differed from Study 2?
 - In Study 1, changes in 3 geometric characteristics of snail shells were studied; but in Study 2, the environment in which the snails lived was studied.
 - In Study 1, changes in 3 geometric characteristics of snail shells were studied; but in Study 2, the variation in the pattern of stripes on snail shells was studied.
 - In Study 1, the variation in the pattern of stripes on snail shells was studied; but in Study 2, the environment in which the snails lived was studied.
 - In Study 1, the variation in the pattern of stripes on snail shells was studied; but in Study 2, changes in 3 geometric characteristics of snail shells were studied.
- Based on the description of the procedures in the studies, the scientists had to extract complete and intact fossil snail shells instead of fossil shell fragments from the rock layers in order to:
 - describe the colors of a living snail's shell.
 - make accurate measurements of each shell's dimensions.
 - determine the present-day climate of Bonin Island.
 - determine the rock layers' ages using radiocarbon dating.
- Based on the description of Study 1, which of the following figures most likely represents the appearance of the B, L, B, B stripe pattern on a snail shell?
 -
 -
 -
 -
- According to the results of Study 2, how do the average number of whorls and the average shell diameter for living Bonin Island snails compare to those of fossil Bonin Island snails with an age of 40,000 years? For the living snails:
 - both the average number of whorls and the average shell diameter are larger.
 - both the average number of whorls and the average shell diameter are smaller.
 - the average number of whorls is larger and the average shell diameter is smaller.
 - the average number of whorls is smaller and the average shell diameter is larger.

Passage II

Most fruit flies have red eyes; some have white eyes. One gene that affects eye color is located on the X chromosome. Female offspring receive 1 X chromosome from both their mother and their father, so they have 2 copies of this gene. Male offspring receive 1 X chromosome from their mother and no X chromosomes from their father (instead, they receive 1 Y chromosome from their father), so they have 1 copy of this gene.

There are 2 *alleles* (forms) of this gene: the red-eye allele (X^R) and the white-eye allele (X^r). X^R is *dominant* to X^r ; and X^r is *recessive* to X^R . This means an individual with 1 X^R and 1 X^r will be red-eyed.

Genotype refers to the combination of alleles an individual has at a gene. The table lists each genotype and the corresponding eye color.

Genotype	Eye color
$X^R X^R$	red
$X^R X^r$	red
$X^r X^r$	white
$X^R Y$	red
$X^r Y$	white

Students did 4 experiments to investigate the inheritance of eye color in fruit flies. In each experiment, 50% of the offspring were females and 50% were males.

Experiment 1

Fifty red-eyed females with Genotype $X^R X^R$ were mated with 50 white-eyed males. The results were as follows:

Females: 100% red-eyed.
Males: 100% red-eyed.

Experiment 2

Fifty female offspring and 50 male offspring from Experiment 1 were mated together. The results were as follows:

Females: 100% red-eyed.
Males: 50% red-eyed; 50% white-eyed.

Experiment 3

Fifty female offspring from Experiment 1 were mated with 50 white-eyed males. The results were as follows:

Females: 50% red-eyed; 50% white-eyed.
Males: 50% red-eyed; 50% white-eyed.

Experiment 4

Fifty white-eyed female offspring from Experiment 3 were mated with 50 red-eyed males. The results were as follows:

Females: 100% red-eyed.
Males: 100% white-eyed.

7. To produce only white-eyed offspring, one would mate fruit flies with which of the following sets of genotypes?
 - A. $X^r X^r$ and $X^r Y$
 - B. $X^R X^r$ and $X^r Y$
 - C. $X^r X^r$ and $X^R Y$
 - D. $X^R X^r$ and $X^R Y$
8. A white-eyed female has which of the following genotypes?
 - F. $X^r X^r$
 - G. $X^R X^r$
 - H. $X^r Y$
 - J. $X^R Y$
9. Some or all of the male offspring had 1 white-eye allele in Experiments:
 - A. 2 and 4 only.
 - B. 3 and 4 only.
 - C. 1, 2, and 3 only.
 - D. 2, 3, and 4 only.
10. Suppose that 1,000 offspring were produced in Experiment 3. Based on the results, the number of red-eyed male offspring produced in Experiment 3 would most likely have been closest to:
 - F. 0.
 - G. 250.
 - H. 500.
 - J. 750.

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11. The ratio of red-eyed offspring to white-eyed offspring produced in Experiment 2 was:

- A. 1:0.
- B. 1:1.
- C. 2:1.
- D. 3:1.

12. In Experiment 3, the parents most likely had which of the following genotypes?

- F. X^rX^r and X^RY
- G. X^RX^R and X^rY
- H. X^RX^r and X^RY
- J. X^RX^r and X^rY

Passage III

The carburetors in gasoline engines have a constriction called a *venturi*, where gasoline and air are mixed (see Figure 1).

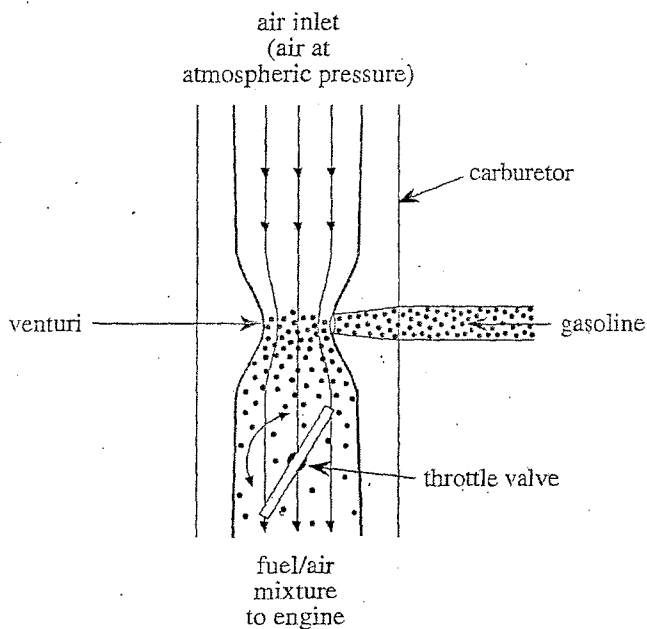


Figure 1

Because the diameter of the venturi is less than the diameter of the air inlet, air moves faster through the venturi than through the air inlet. As a result, the pressure in the venturi is less than atmospheric pressure, and gasoline is drawn into the venturi. Moreover, the faster air moves through the venturi, the lower the pressure in the venturi, and the greater the rate at which gasoline is drawn into the venturi.

The *percent change in pressure* ($\% \Delta P$) can be defined as:

$$\% \Delta P = \frac{\text{atmospheric pressure} - \text{pressure in a venturi}}{\text{atmospheric pressure}} \times 100$$

The $\% \Delta P$ of carburetors was determined under various conditions in the following experiments.

Experiment 1

A carburetor was mounted on an engine with an *air volume capacity* (the volume of air drawn into the engine during each piston stroke) of 0.55 liter (L). The $\% \Delta P$ of the carburetor was found at various engine speeds. The results are given in Table 1.

Table 1	
Engine speed (rpm)	$\% \Delta P$
1,000	0.3
3,000	2.4
5,000	6.7
7,000	13.1

Experiment 2

The engine used in Experiment 1 was fitted with carburetors having the same air inlet diameter as in Experiment 1 but different venturi radii. The $\% \Delta P$ for each carburetor was measured at an engine speed of 3,000 rpm (see Table 2).

Table 2	
Venturi radius (cm)	$\% \Delta P$
0.8	12.2
1.0	5.0
1.2	2.4
1.4	1.3
1.6	0.8

Experiment 3

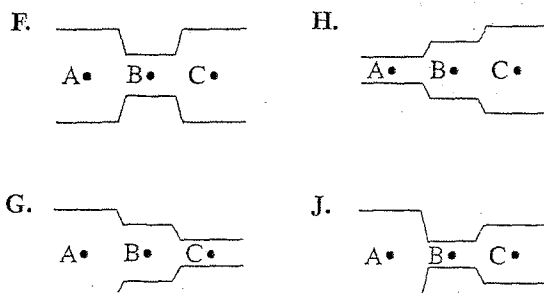
The carburetor used in Experiment 1 was fitted to engines having various air volume capacities. The $\% \Delta P$ for each engine was measured at an engine speed of 3,000 rpm (see Table 3).

Table 3	
Air volume capacity (L)	$\% \Delta P$
0.45	1.6
0.50	2.0
0.55	2.4
0.60	2.9
0.65	3.4

13. Under the conditions described for Experiment 3, a $\% \Delta P$ of 2.2 would most likely be obtained if the air volume capacity equaled:
- 0.45 L.
 - 0.53 L.
 - 0.58 L.
 - 0.63 L.
14. Based on the results of Experiment 1, if engine speed had been less than 1,000 rpm, $\% \Delta P$ would most likely have been:
- less than 0.3.
 - between 0.3 and 2.4.
 - between 2.4 and 6.7.
 - greater than 6.7.
15. Which of the following is the most likely explanation for the results of Experiment 1? As engine speed increases, $\% \Delta P$:
- increases, because air speed through the venturi increases.
 - increases, because air speed through the venturi decreases.
 - decreases, because air speed through the venturi increases.
 - decreases, because air speed through the venturi decreases.
16. Consider a horizontal pipe of varying diameter that is filled with helium gas flowing through the pipe. The speed of the gas in meters per second (m/sec) and the pressure of the gas in atmospheres (atm) at 3 locations in the pipe are given in the following table.

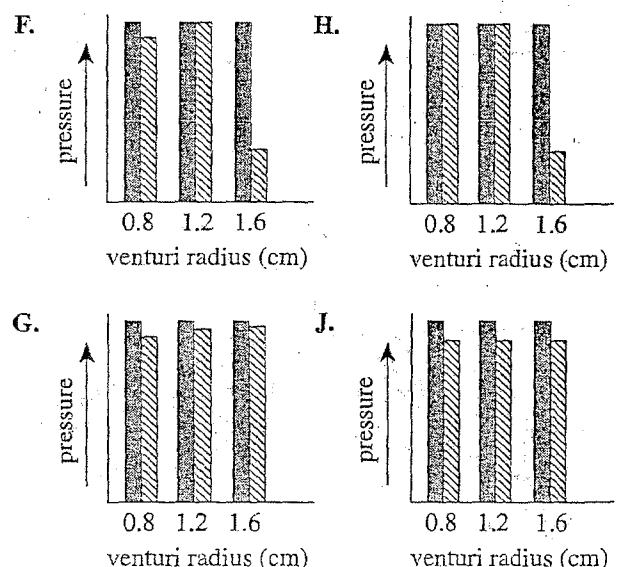
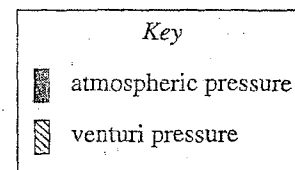
Location	Speed (m/sec)	Pressure (atm)
A	2.2	3.1
B	6.7	2.9
C	5.0	3.0

Based on the information in the passage about air flow in a carburetor, which of the following drawings most likely represents the changes in the pipe's diameter?



17. Based on the results of Experiments 1 and 2, what was the venturi radius of the carburetor used in Experiment 3?
- 1.0 cm
 - 1.2 cm
 - 1.4 cm
 - 1.6 cm

18. For the carburetors used in Experiment 2 with venturi radii of 0.8 cm, 1.2 cm, and 1.6 cm, which of the following graphs best displays the atmospheric pressure at each air inlet and the pressure measured in each venturi?



Passage IV

When substances are heated, they typically *expand* (increase in volume). Under certain circumstances, some substances will *contract* (decrease in volume) when heated. Table 1 shows the change in volume that occurred when 1.0000 L samples of solids at different initial temperatures were heated. Tables 2 and 3 show the results for similar trials using 1.0000 L samples of liquids and gases, respectively.

(Note: No phase changes occurred in any of the trials, and the gas samples were heated at constant pressure.)

Table 1

Solid	Initial temperature of solid (°C)	Final temperature of solid (°C)	Change in volume (L)
Lead	-6	4	0.0009
Lead	4	14	0.0009
Lead	14	24	0.0009
Water	-16	-6	0.0015
Water	-6	0	0.0008
Silver	-6	4	0.0006
Silver	4	14	0.0006
Silver	14	24	0.0006

Table 2

Liquid	Initial temperature of liquid (°C)	Final temperature of liquid (°C)	Change in volume (L)
Ethanol	-6	4	0.0075
Ethanol	4	14	0.0075
Ethanol	14	24	0.0075
Water	0	4	-0.0050
Water	4	14	0.0013
Water	14	24	0.0020
Mercury	-6	4	0.0018
Mercury	4	14	0.0018
Mercury	14	24	0.0018

Table 3

Gas	Initial temperature of gas (°C)	Final temperature of gas (°C)	Change in volume (L)
Helium	-6	4	0.0374
Helium	4	14	0.0361
Helium	14	24	0.0348
Oxygen	-6	4	0.0374
Oxygen	4	14	0.0361
Oxygen	14	24	0.0348
Methane	-6	4	0.0374
Methane	14	24	0.0348

19. Which of the following substances listed in Tables 1–3 expanded by the same amount each time it was heated, regardless of its initial temperature?
- A. Solid water
B. Liquid water
C. Mercury
D. Oxygen
20. Based on Table 1, if a 1.0000 L sample of lead at 24°C is *cooled* to 14°C, the volume will most likely:
- F. decrease by 0.0009 L.
G. decrease by 0.0027 L.
H. increase by 0.0009 L.
J. increase by 0.0027 L.
21. A scientist claimed that whenever water is heated, it will expand. Based on Tables 1 and 2, is his claim valid?
- A. Yes; in every trial that water was heated, the change in volume was positive.
B. No; in every trial that water was heated, the change in volume was negative.
C. Yes; when water was heated from -16°C to 24°C, the change in volume was positive.
D. No; when water was heated from 0°C to 4°C, the change in volume was negative.

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22. Which of the following is a plausible explanation for the change in volume observed for the lead samples listed in Table 1 ? As the lead samples were heated, the lead atoms in the sample moved:

- F. closer together, causing the sample to increase in volume.
- G. closer together, causing the sample to decrease in volume.
- H. farther apart, causing the sample to increase in volume.
- J. farther apart, causing the sample to decrease in volume.

23. Based on Table 3, if 1.0000 L of helium at 24°C is heated to 34°C at constant pressure, the change in volume will most likely be closest to which of the following?

- A. 0.0335 L
- B. 0.0348 L
- C. 0.0361 L
- D. 0.0374 L

Passage V

As shown in Figure 1, the temperatures of the atmospheres of Venus, Earth, and Mars vary with altitude. Figure 1 also shows how the atmospheric temperature for Earth and Venus varies between day and night and shows the layers of Earth's atmosphere. Table 1 gives the composition of the atmosphere on each planet, both as relative abundance and as the partial pressure, in bars, exerted by each gas at the planet's surface. The total pressure of the gases in Earth's atmosphere at Earth's surface is approximately 1 bar. Table 2 gives what average surface temperatures would be on the 3 planets without greenhouse warming.

Table 2	
Planet	Average surface temperature without greenhouse warming (kelvin)
Venus	280
Earth	250
Mars	220

Figure and tables adapted from J. Kelly Beatty, Carolyn Collins Petersen, and Andrew Chaikin, *The New Solar System*, 4th ed. ©1999 by Cambridge University Press.

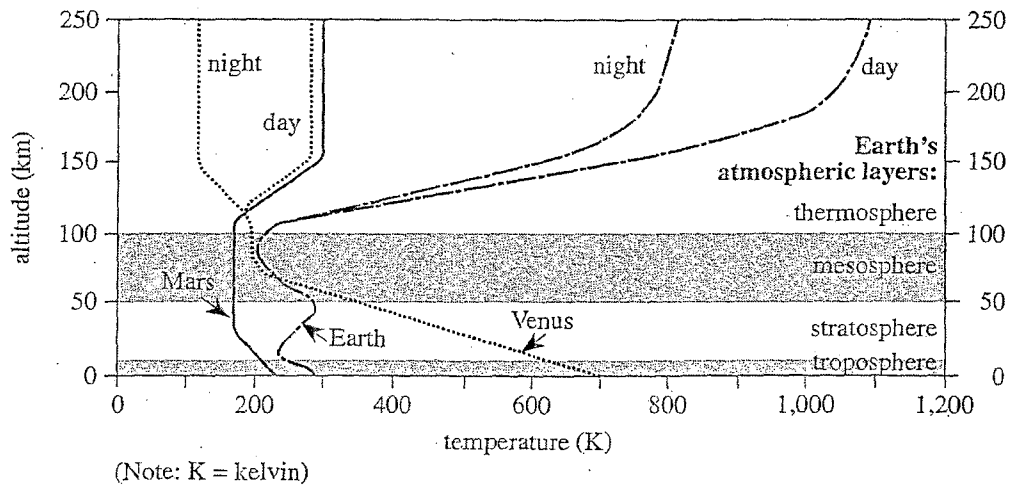


Figure 1

Table 1						
Gas	Relative abundance (%)			Partial pressure exerted by each gas at the planet's surface (bars)		
	Venus	Earth	Mars	Venus	Earth	Mars
CO ₂	96	0.035	95	86.4	0.00035	0.0062
N ₂	3.5	77	2.7	3.2	0.78	0.00018
O ₂	0	21	0	0	0.21	0
Ar	0.007	0.94	1.6	0.0063	0.0095	0.0001
H ₂ O	0.01	1	0.006	0.009	0.01	0.00000039

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24. According to Figure 1, the temperature of Mars's atmosphere changes the most between:
- F. 25 km and 50 km.
 - G. 75 km and 100 km.
 - H. 125 km and 150 km.
 - J. 200 km and 225 km.
25. Considering only the gases listed in Table 1, which gas simultaneously composes the largest percent of Mars's atmosphere and the smallest percent of Earth's atmosphere?
- A. CO_2
 - B. O_2
 - C. N_2
 - D. Ar
26. According to Figure 1, the temperature of Venus's atmosphere remains the same as the altitude increases from:
- F. 0 km to 50 km.
 - G. 50 km to 100 km.
 - H. 100 km to 150 km.
 - J. 150 km to 200 km.
27. Which of the following statements about O_2 in the atmospheres of the 3 planets is supported by the data in Table 1?
- A. The atmospheres of Venus and Earth contain equal quantities of O_2 .
 - B. The atmospheres of Earth and Mars contain equal quantities of O_2 .
 - C. The atmospheres of Venus and Earth contain no O_2 .
 - D. The atmospheres of Venus and Mars contain no O_2 .
28. Based on Table 2, the average temperature at Venus's surface *without* greenhouse warming would be how many degrees cooler than the temperature given in Figure 1?
- F. 280 K
 - G. 360 K
 - H. 420 K
 - J. 700 K

Passage VI

Nuclear fusion is the combining of the *nuclei* (central cores) of atoms and can be used to produce energy. Different nuclei contain various numbers of *protons* (particles with a single positive electrical charge) and *neutrons* (uncharged particles).

A *containment vessel* holds a mixture of *deuterium* nuclei (containing 1 proton and 1 neutron) and *tritium* nuclei (containing 1 proton and 2 neutrons). (The density, n , is the number of nuclei/cm³; higher densities increase the frequency of collisions between nuclei.) The nuclei collide, fuse, and emit high amounts of energy. Because like-charged particles repel each other, getting two nuclei close enough to each other to fuse requires that the nuclei be accelerated to high speeds, which consumes energy. Moreover, energy is consumed in preventing the nuclei from striking the walls of the containment vessel, where the nuclei would lose speed.

To produce more energy from fusion than is consumed, the temperature must be maintained above 4×10^7 K for a minimum time, τ , and the product of n and τ must exceed 10^{14} sec/cm³ (a requirement called *Lawson's criterion*).

Two methods for *confinement* (preventing nuclei from colliding with vessel walls) and for producing nuclear fusion are discussed below.

The Magnetic Confinement Method

A magnetic force field compresses a gas of tritium and deuterium nuclei. Because excessive gas densities will cause the magnetic field to buckle, allowing the gas to escape confinement and to strike the walls of the containment vessel, a maximum n of about 10^{14} nuclei/cm³ is allowed. Therefore, in order to satisfy Lawson's criterion, a temperature of approximately 10^8 K is maintained for about 1 sec.

The Laser Confinement Method

A pellet containing deuterium and tritium atoms is subjected to pulses of laser light, producing a gas with a density of about 10^{23} nuclei/cm³. The laser pulses are shaped to compress and heat the gas to approximately 10^8 K for about 10^{-9} sec. Unfortunately, some of the laser energy is reflected rather than absorbed by the gas.

At present, the ratio of power (energy per unit time) produced to the power consumed by the Magnetic Confinement Method is higher than that by the Laser Confinement Method. However, because of the very high density of nuclei around the pellet, the power produced by the Laser Confinement Method can be higher than that of magnetic confinement.

29. Both confinement methods consume large amounts of energy in order to:

- A. cause fusion between nuclei.
- B. increase the distances between nuclei.
- C. decrease the speed of the nuclei.
- D. decrease the charges on the nuclei.

30. According to the passage, for the Magnetic Confinement Method approximately what is the value of τ ?

- F. 10^{-10} sec
- G. 10^{-5} sec
- H. 1 sec
- J. 10 sec

31. For both confinement methods, fusing deuterium nuclei is difficult because the nuclei have identical:

- A. charges.
- B. masses.
- C. speeds.
- D. weights.

32. To increase the probability that nuclei will collide, a goal of both methods is to:

- F. decrease the temperature inside the containment vessel.
- G. decrease the density of nuclei in the containment vessel.
- H. increase the density of nuclei in the containment vessel.
- J. increase the number of neutrons in the nuclei.

33. A comparison of the densities of the deuterium/tritium gas for the two methods shows that the:

- A. Magnetic Confinement Method typically produces higher densities.
- B. Laser Confinement Method typically produces higher densities.
- C. Laser Confinement Method never produces higher densities.
- D. two methods usually produce equal densities.

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34. In the Laser Confinement Method, laser light is used to:

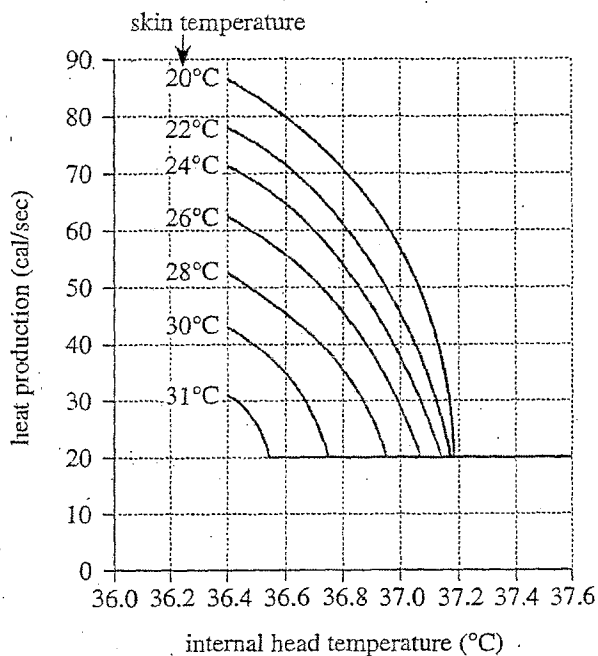
- F. cool the deuterium and tritium nuclei.
- G. exert pressure on the deuterium and tritium nuclei.
- H. illuminate the walls of the containment vessel.
- J. heat the walls of the containment vessel.

35. For a new method of confinement, the product of n and τ is about 10^{13} sec/cm³. Which of the following changes would be most likely to bring the new method into compliance with Lawson's criterion?

- A. A large decrease in n and a small increase in τ
- B. A small increase in n and a large decrease in τ
- C. Large decreases in both n and τ
- D. Large increases in both n and τ

Passage VII

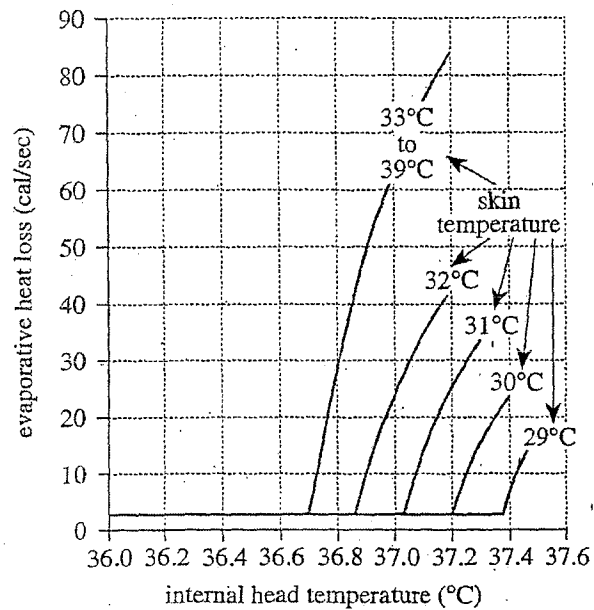
Figure 1 shows how, on average, internal head temperature and skin temperature affect heat production (in calories per second, cal/sec) in humans.



(Note: For each skin temperature, the range for internal head temperature begins at 36.4°C and ends at 37.6°C.)

Figure 1

Figure 2 shows how, on average, internal head temperature and skin temperature affect *evaporative heat loss* (heat lost through the evaporation of perspiration) in humans.



(Note: For each skin temperature, the range for internal head temperature begins at 36.0°C.)

Figure 2

Figures adapted from Arthur C. Guyton, M.D., *Textbook of Medical Physiology*. ©1991 by W. B. Saunders Company.

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36. According to Figure 2, if internal head temperature is 37.0°C and evaporative heat loss is 14 cal/sec , skin temperature is most likely within which of the following ranges?

F. 29°C – 30°C
G. 30°C – 31°C
H. 31°C – 32°C
J. 32°C – 33°C

37. According to Figure 1, at a skin temperature of 20°C , which of the following changes in internal head temperature will cause the greatest decrease in heat production?

A. From 36.6°C to 36.8°C
B. From 36.8°C to 37.0°C
C. From 37.0°C to 37.2°C
D. From 37.2°C to 37.4°C

38. According to Figure 2, at a skin temperature of 32°C , as internal head temperature increases from 36.2°C to 37.0°C , evaporative heat loss:

F. remains constant, then decreases.
G. remains constant, then increases.
H. increases, then decreases.
J. decreases, then increases.

39. According to Figure 1, which of the following combinations of internal head temperature and skin temperature results in the greatest heat production?

	Internal head temperature ($^{\circ}\text{C}$)	Skin temperature ($^{\circ}\text{C}$)
A.	36.6	22
B.	36.6	28
C.	37.0	22
D.	37.0	28

40. Based on Figure 1, if an individual has an internal head temperature of 36.8°C and a skin temperature of 29°C , the individual's heat production is most likely closest to which of the following?

F. 30 cal/sec
G. 40 cal/sec
H. 50 cal/sec
J. 60 cal/sec

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

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English				Mathematics				Reading		Science	
1	C	36	J	1	A	31	C	1	B	1	C
2	J	37	C	2	J	32	J	2	J	2	G
3	B	38	F	3	E	33	D	3	C	3	D
4	G	39	C	4	J	34	H	4	F	4	G
5	A	40	G	5	D	35	D	5	D	5	B
6	J	41	C	6	H	36	G	6	G	6	H
7	B	42	F	7	B	37	C	7	A	7	A
8	G	43	D	8	H	38	K	8	J	8	F
9	B	44	G	9	C	39	A	9	C	9	D
10	J	45	A	10	H	40	H	10	H	10	G
11	C	46	G	11	B	41	A	11	C	11	D
12	J	47	B	12	J	42	K	12	F	12	J
13	B	48	G	13	E	43	A	13	C	13	B
14	J	49	A	14	G	44	K	14	F	14	F
15	A	50	G	15	B	45	E	15	A	15	A
16	F	51	C	16	K	46	G	16	H	16	J
17	D	52	H	17	E	47	E	17	D	17	B
18	J	53	C	18	H	48	G	18	G	18	G
19	A	54	J	19	E	49	E	19	D	19	C
20	F	55	A	20	J	50	F	20	J	20	F
21	D	56	H	21	B	51	E	21	C	21	D
22	H	57	A	22	F	52	G	22	G	22	H
23	D	58	G	23	C	53	D	23	D	23	A
24	H	59	C	24	J	54	K	24	G	24	H
25	C	60	F	25	A	55	C	25	A	25	A
26	F	61	B	26	H	56	J	26	H	26	J
27	B	62	H	27	A	57	A	27	D	27	D
28	F	63	D	28	G	58	G	28	G	28	H
29	C	64	F	29	A	59	A	29	D	29	A
30	G	65	C	30	F	60	H	30	J	30	H
31	C	66	F					31	A	31	A
32	J	67	D					32	G	32	H
33	B	68	H					33	C	33	B
34	H	69	D					34	H	34	G
35	A	70	F					35	D	35	D
		71	D					36	F	36	H
		72	F					37	D	37	C
		73	A					38	F	38	G
		74	J					39	C	39	A
		75	A					40	G	40	F